



DEN

EXECUTIVE REFERENCE DATA NATIONAL ENERGY MANAGEMENT

NATIONAL ENERGY COUNCIL
2015

PREFACE

Once again we would like to convey some information about the situation energy management in Indonesia, as the publication in the year 2015, that contains energy management data by year 2014, such as socio-economic, energy indicator, the management each type of energy covering oil and gas, coal, renewable energy, and electricity.

The Indonesian economic growth which is quite high made Indonesia as one of the economic potential which are still strong in Asia. Moreover the past few years with the condition of the global crisis that sweeping the world, Indonesian economic development can grow on the high level of domestic energy consumption. The economic growth of Indonesia at an average annual rate of 5.78% during the period 2012 through 2013, has triggered the increase of domestic energy demand by an average of 4.4% per year. By 2030 the energy demand is estimated to be three times the demand level in 2010, while the increase in energy production in 2030 is forecast to rise at twice that of the year 2010.

This information is expected to provide an overview on the future of energy management that can address the challenges in meeting national energy needs.

We would like to thank and appreciate all parties, for their contribution to the preparation of this booklet.

Jakarta, April 2015

Secretary General of
National Energy Council

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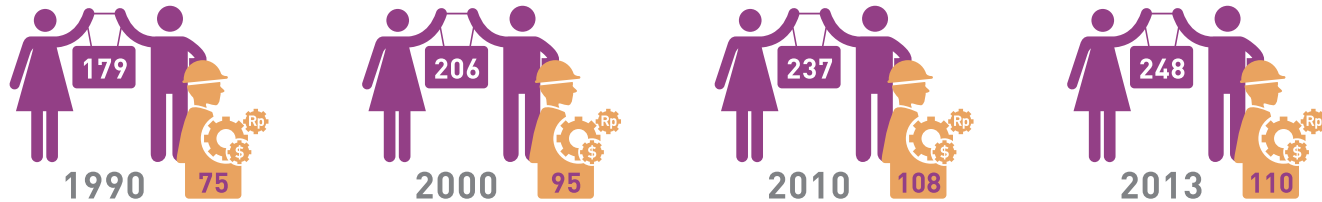


SOCIO - ECONOMIC

POPULATION 2013

POPULATION AND ECONOMICALLY ACTIVE

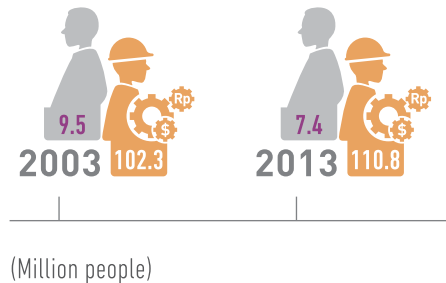
(Million people)



In period 1990 – 2013 population of Indonesia growth 1.4% per year.

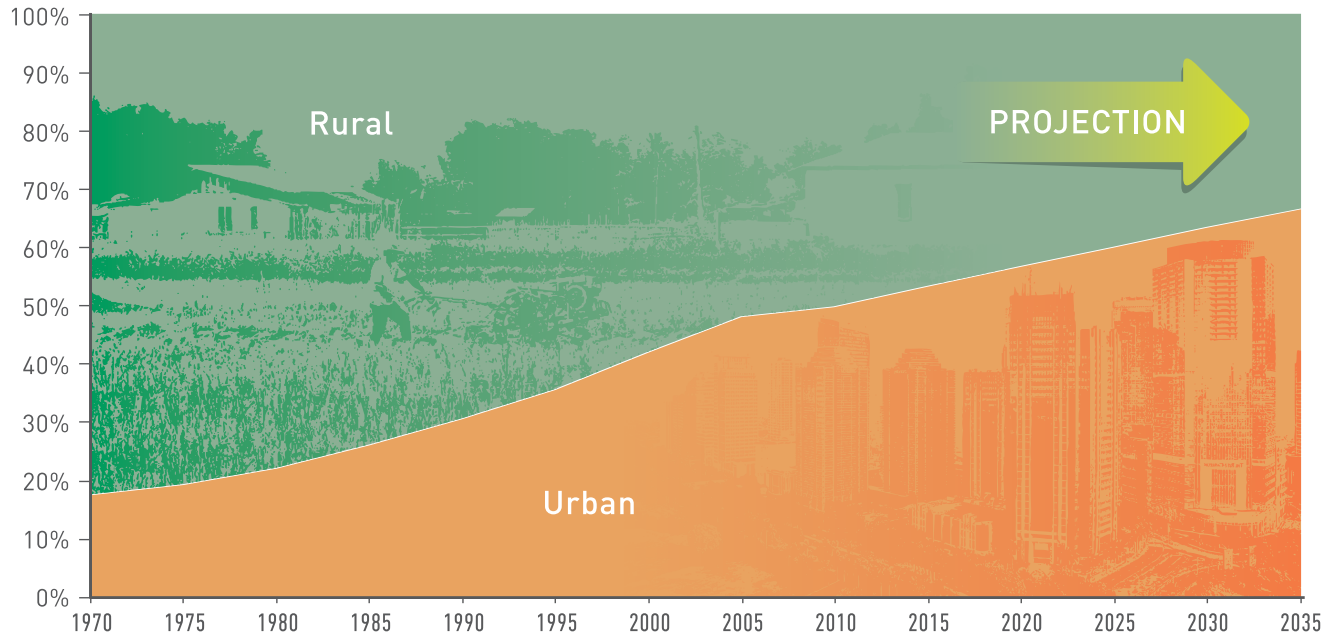


In 2013, 44% is economically active.



Indonesia is facing “demographical bonus”, in which productive age (15 – 54 years) is greater 68% compared to the non-productive age. This of course would be beneficial if there is an increase in employment and the improvement of education in Indonesia.

POPULATION IN RURAL AND URBAN AREAS



Source: BPS - Statistics Indonesia 2014, reprocessed by NEC

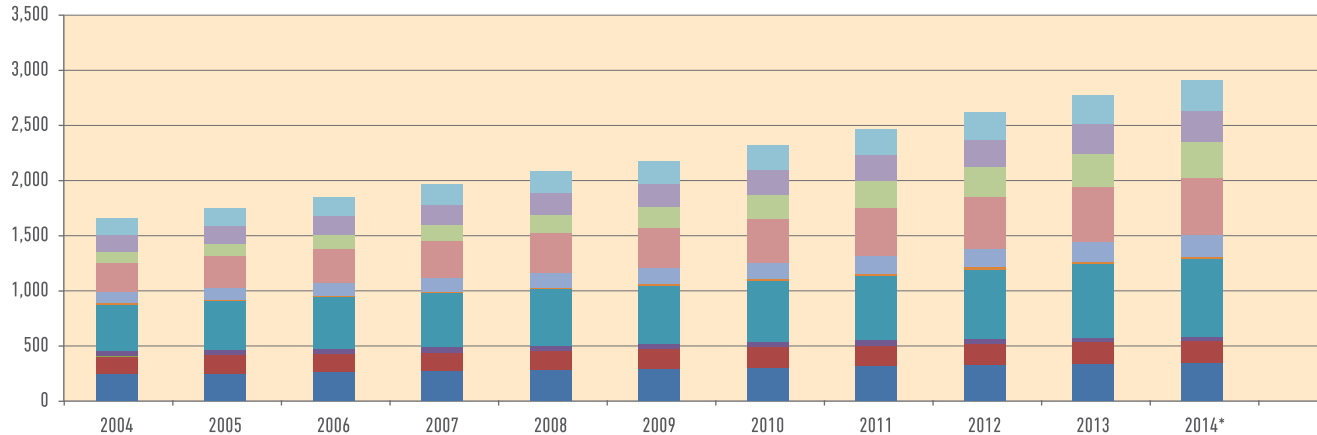
In 1970, number of population in urban area 17.5% increase 48.1% in 2005 (increase an average 2.9 % per year).

In period 2005 – 2013, Urban people increase slowly by an average of 1.2 % per year.

In 2035, the population in the urban areas is estimated/projected to be increased by the composition of 66.6% living in town/city and 33.4% living in villages. The change of a village to become a town has caused the increase in number and density of population, and the economic activity is no longer relies on the agricultural sector, due to the improvement of infrastructure.

GDP PER SECTOR (CONSTANT 2000)

(Trillion IDR)



Source: BPS - Statistics Indonesia 2014, reprocessed by NEC

* Temporary Value

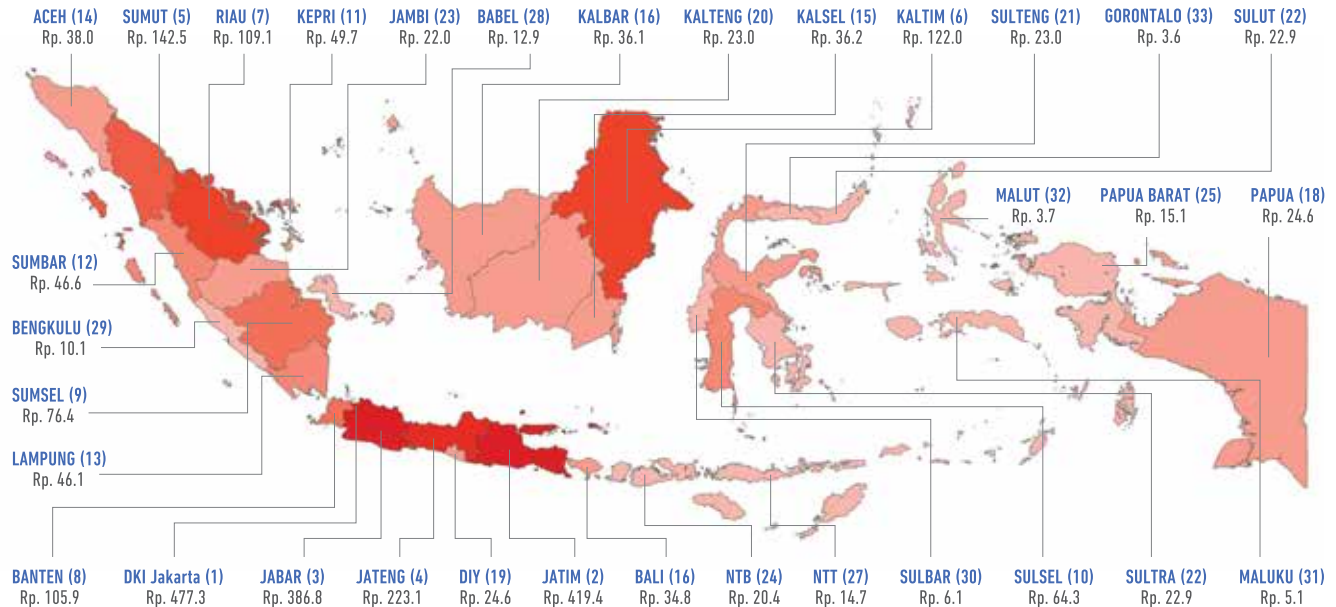
- Services
- Transport and Communication
- Construction
- Non - Oil & Gas Manufacturing Industry
- Mining and Quarrying
- Finance, Real Estate and Business Services
- Trade, Hotel & Restaurants
- Electricity, Gas & Water Supply
- Oil and Gas Manufacturing Industry
- Agriculture, Livestock, Forestry and Fishery

In 2014, GDP of Indonesia is 2,909 trillion rupiahs or increased one and a half times when compared with the year 2004 (growth 5.8 % per year), with the growth occurs in all sectors (the highest growth is in the in transportation and communication sectors about 12.6 % per year). Oil & Gas decreased about 1.7 % and Mining industry growth 2% per year.

In 2014, the share of mining, oil and gas Industry to the Indonesia's total GDP is about 6.7% and 1.5% respectively.

GDP BY REGION

(GDP in Trillion IDR)



Source: BPS - Statistics Indonesia 2014, reprocessed by NEC

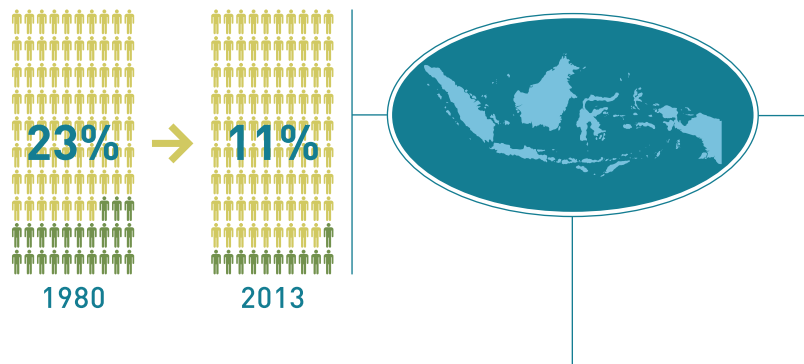


In 2013, DKI Jakarta province have the highest GDP, around IDR 473 trillion with population of 10.1 million, while the province with the lowest GDP is Gorontalo Province namely IDR 3 trillion with population amount 1.1 million people.

SOCIAL AND WELFARE

Poor People

28 million people (11%) of Indonesia's population is poor, of which 8.5% domiciling in the urban area and 14.4% residing in the rural area.

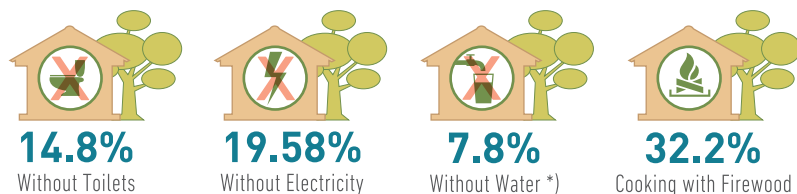


Education

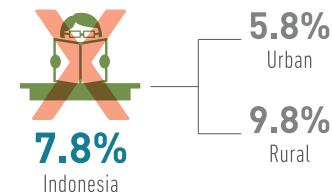
(Thousand)

	 SCHOOL	 STUDENT	 TEACHER
 KINDERGARTEN	70.9	3,612.4	275.1
 ELEMENTARY SCHOOL	146.8	27,583.9	1,550.5
 JUNIOR HIGH SCHOOL	33.6	9,425.3	513.8
 SENIOR HIGH SCHOOL	21.9	8,215.6	440.1
 UNIVERSITY	3.1	6,616.6	192.9

Housing & Environment



Never Attending School



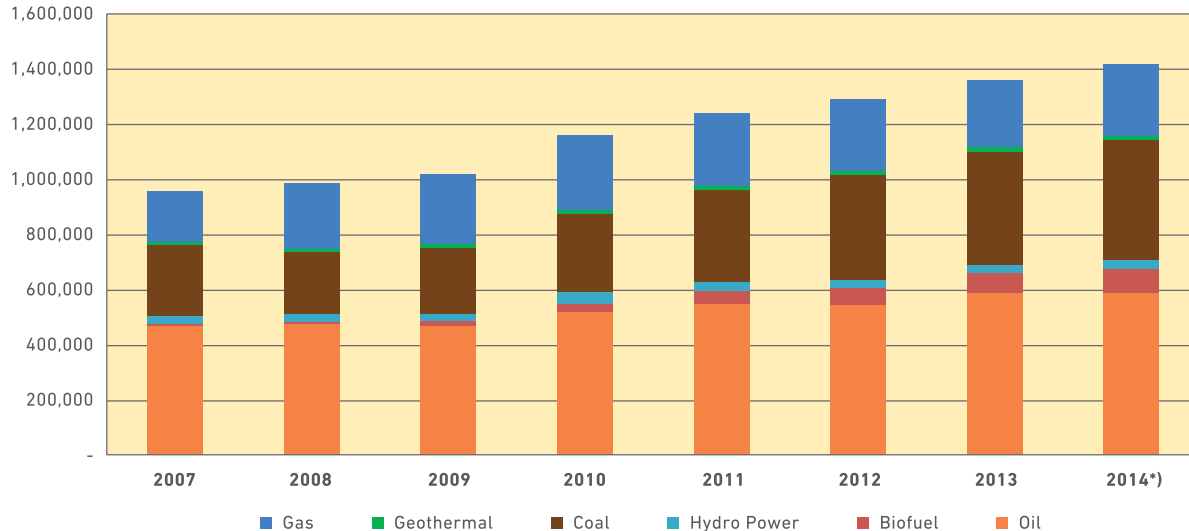
Source: BPS - Statistics Indonesia 2014, Ministry of Education and Culture 2014, reprocessed by NEC

*) Unprotected spring, rivers, and rain water



PRIMARY ENERGY SUPPLY

(Thousand BOE)

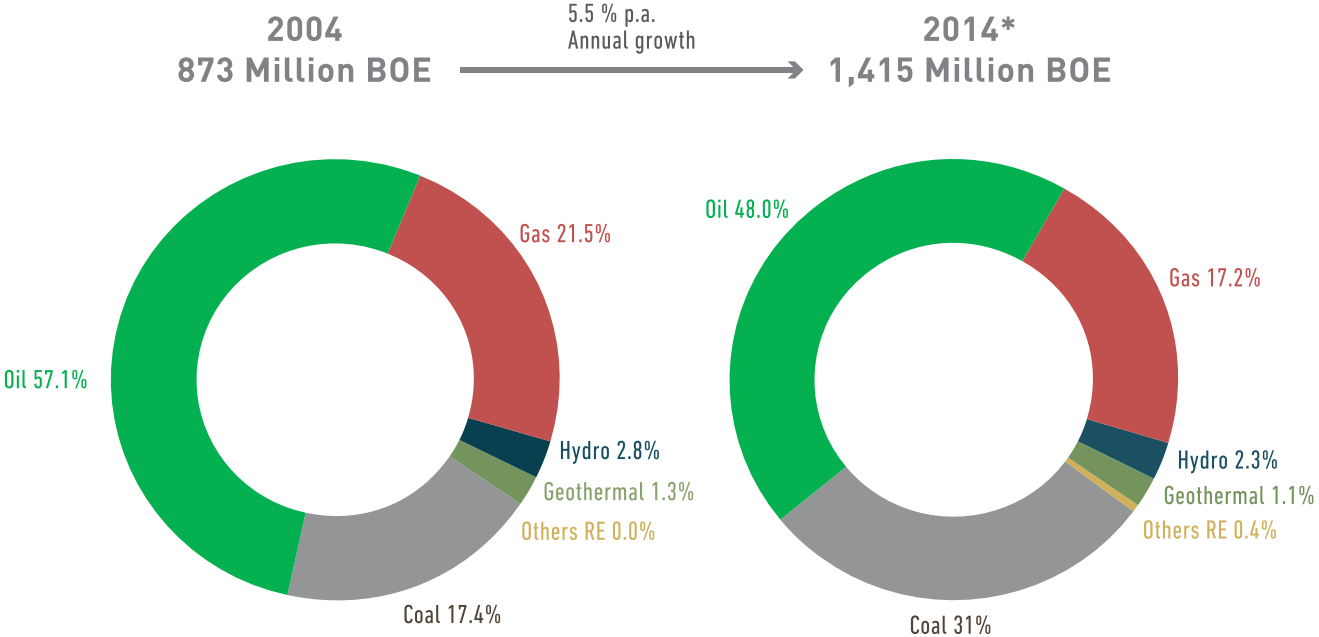


Source: Ministry of Energy and Mineral Resources/MEMR, reprocessed by NEC.

*) temporary 2014 data

Primary energy supply continues to increase with an average of 4.3% per year. The largest energy supply is dominated by petroleum, followed by coal and natural gas.

PRIMARY ENERGY MIX

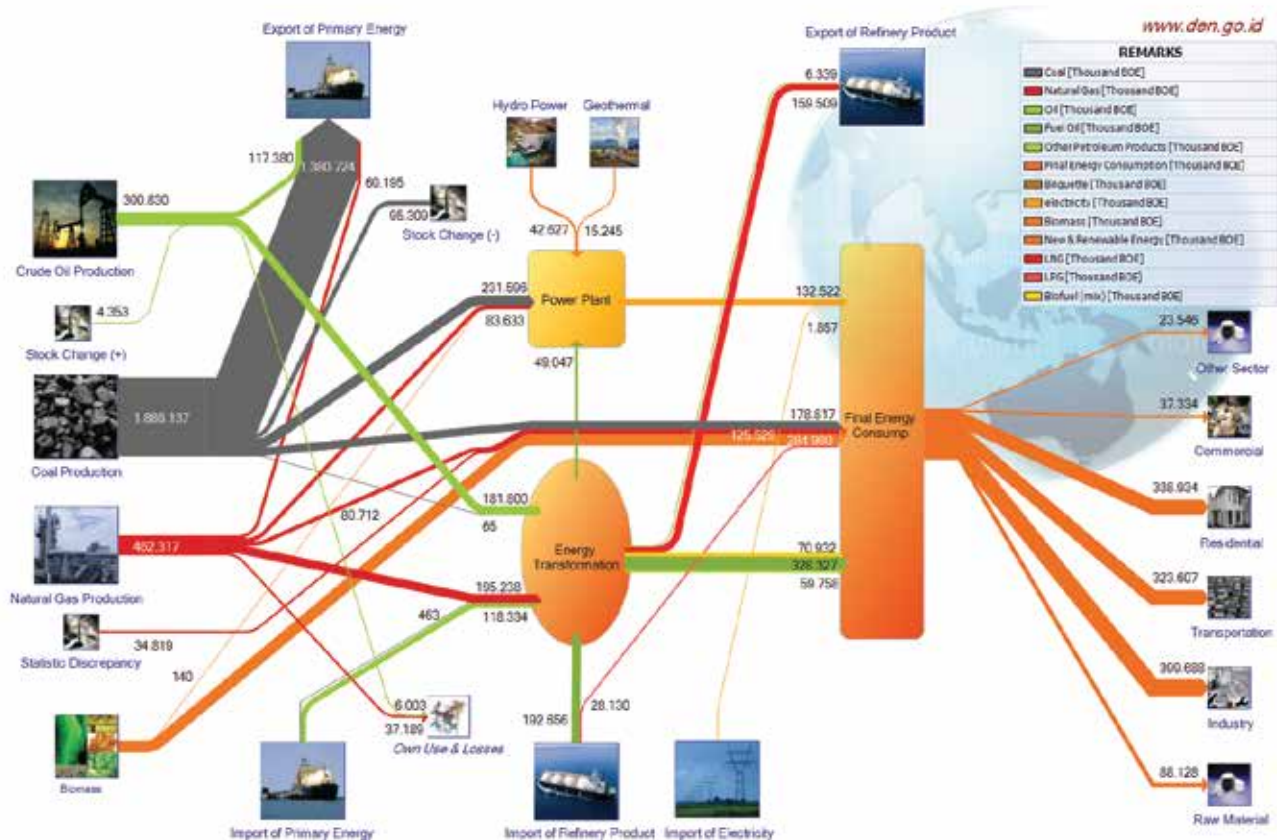


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

*) Temporary data

Total primary energy consumption in the last ten years increased on average by 4.9% per year, oil is still the most energy consumed in the primary energy mix by 48% in 2014.

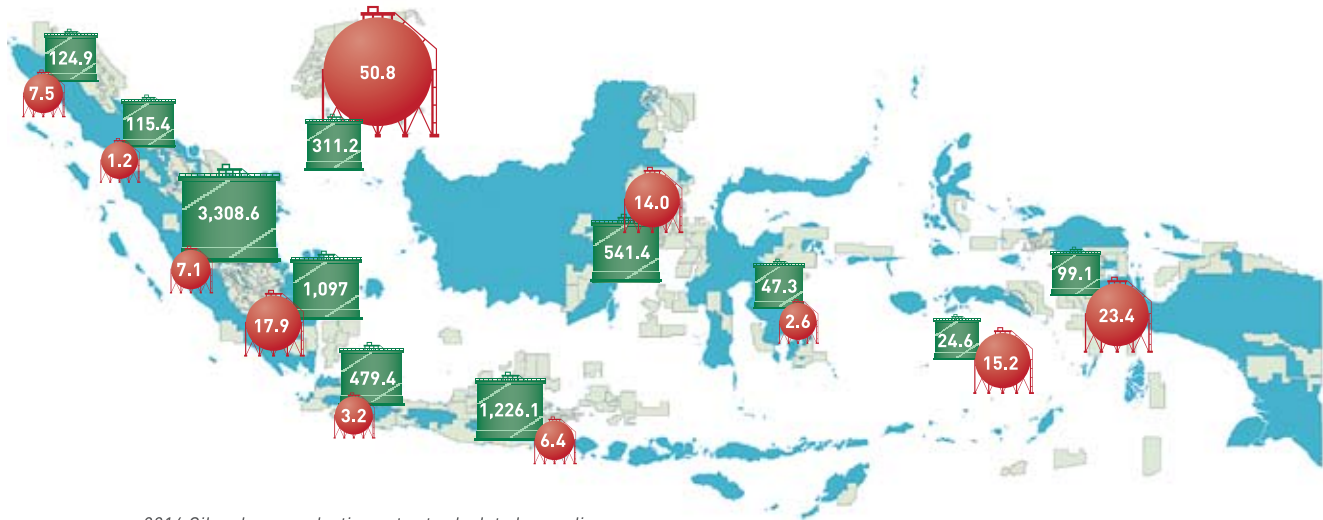
ENERGY FLOW, 2013



An abstract graphic in the top-left corner of the page. It consists of numerous blue cubes and rectangular prisms of varying sizes and orientations. Some are solid blue, while others are semi-transparent, creating a layered, three-dimensional effect. The shapes are scattered across the top-left quadrant, extending towards the center of the page.

3 OIL AND GAS

OIL AND GAS RESERVES



2014 Oil and gas production not yet calculated regarding reserves

OIL REMAINING RESERVE (MMSTB)

PROVEN (P1) = 3,624.3
 POTENTIAL (P2+P3) = 3,750.9
 TOTAL (3P) = 7,375.1

-2.3 %

Compared to 2013

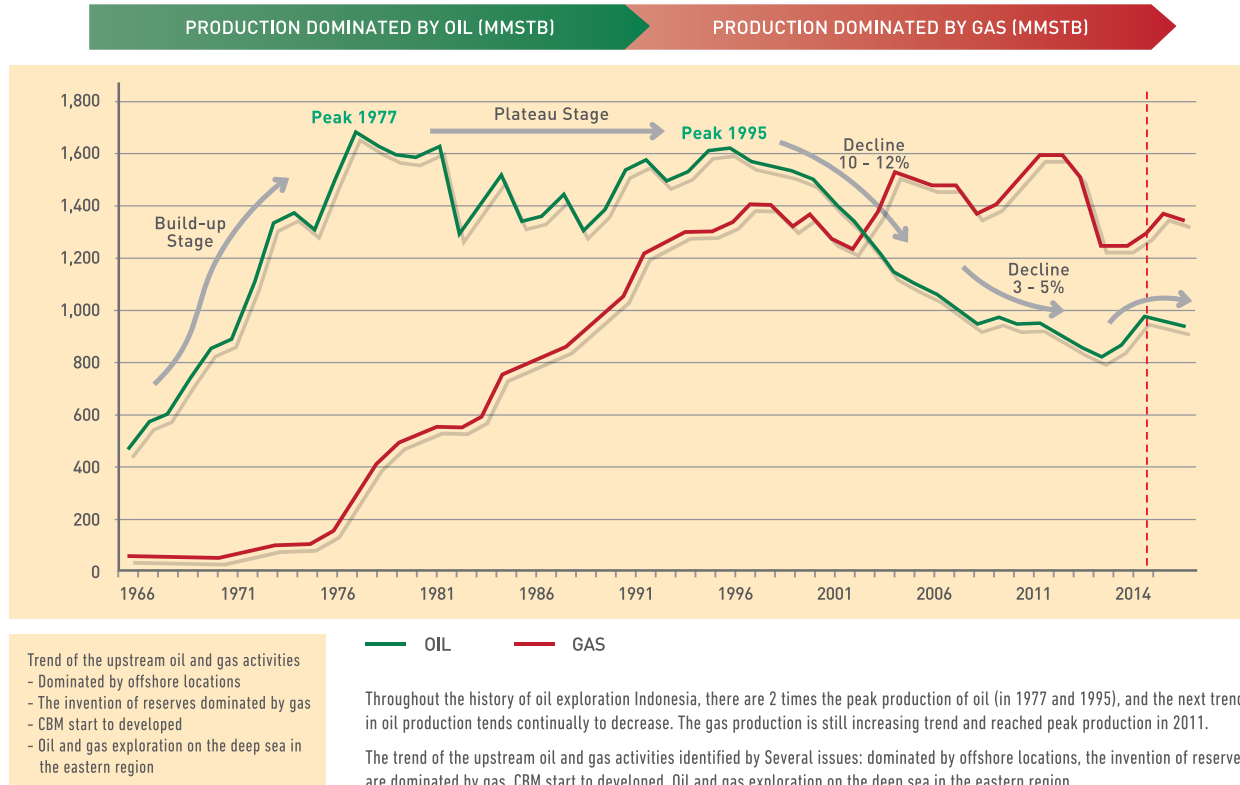
GAS REMAINING RESERVE (TSCF)

PROVEN (P1) = 100.3
 POTENTIAL (P2+P3) = 49.0
 TOTAL (3P) = 149.3

-0.9 %

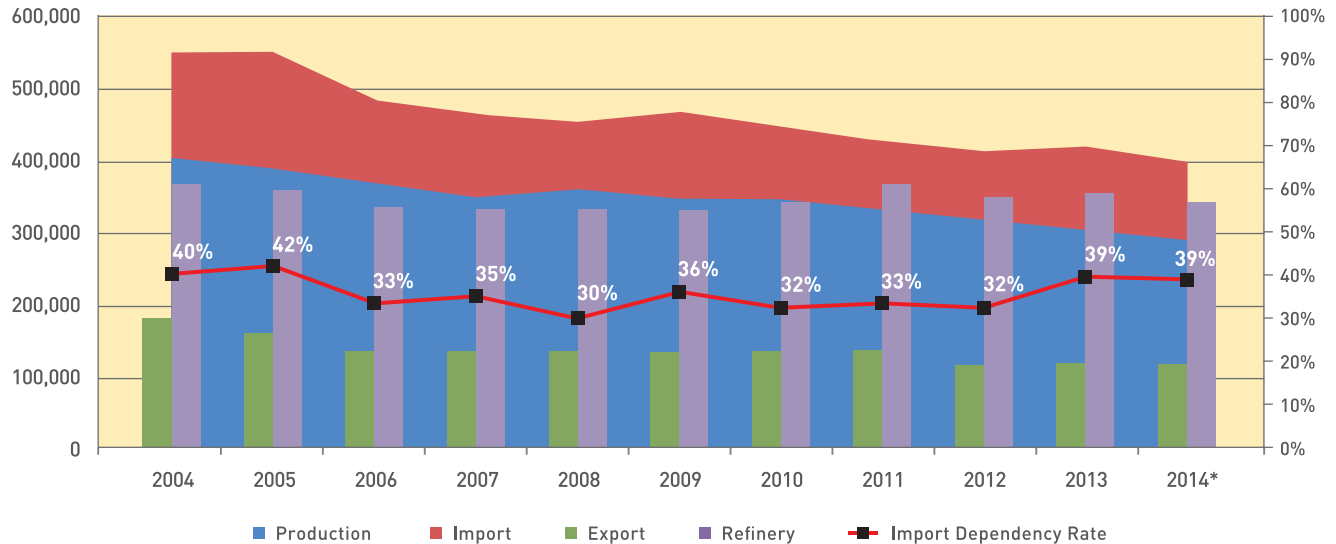
Compared to 2013

OIL AND GAS PRODUCTION PROFILE



CRUDE OIL SUPPLY

(Thousand Bbl)

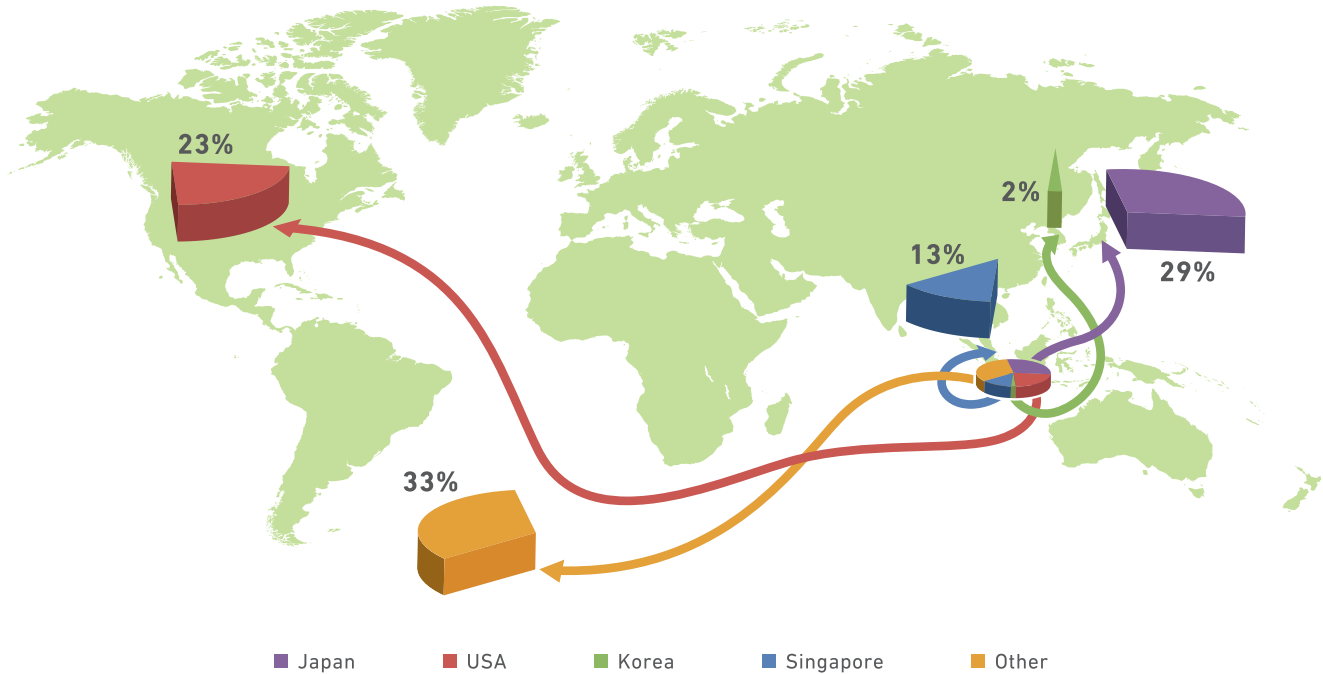


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

* Temporary data

Note : - Discrepancy between production, export, import and domestic demand due to the existing stock
 - The role of imported crude in 2014 was amounted to 32% from the total that goes to the oil refinery

CRUDE OIL EXPORTS

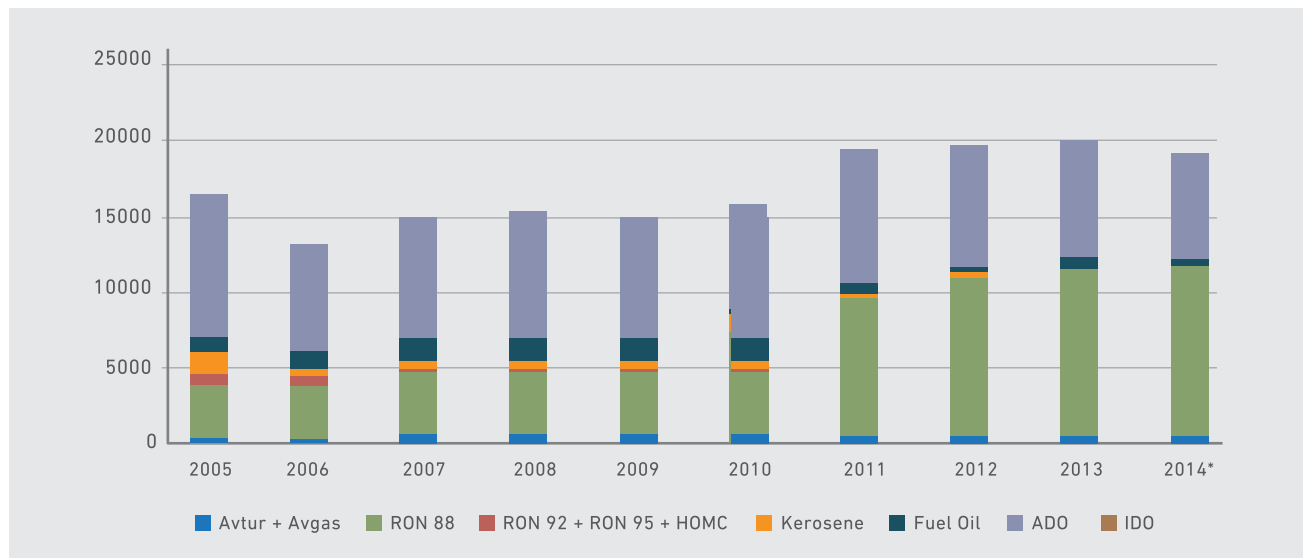


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

In 2014, Indonesia exported crude oil amounted to 116.95 million barrels

FUEL IMPORTS

(Thousand KL)



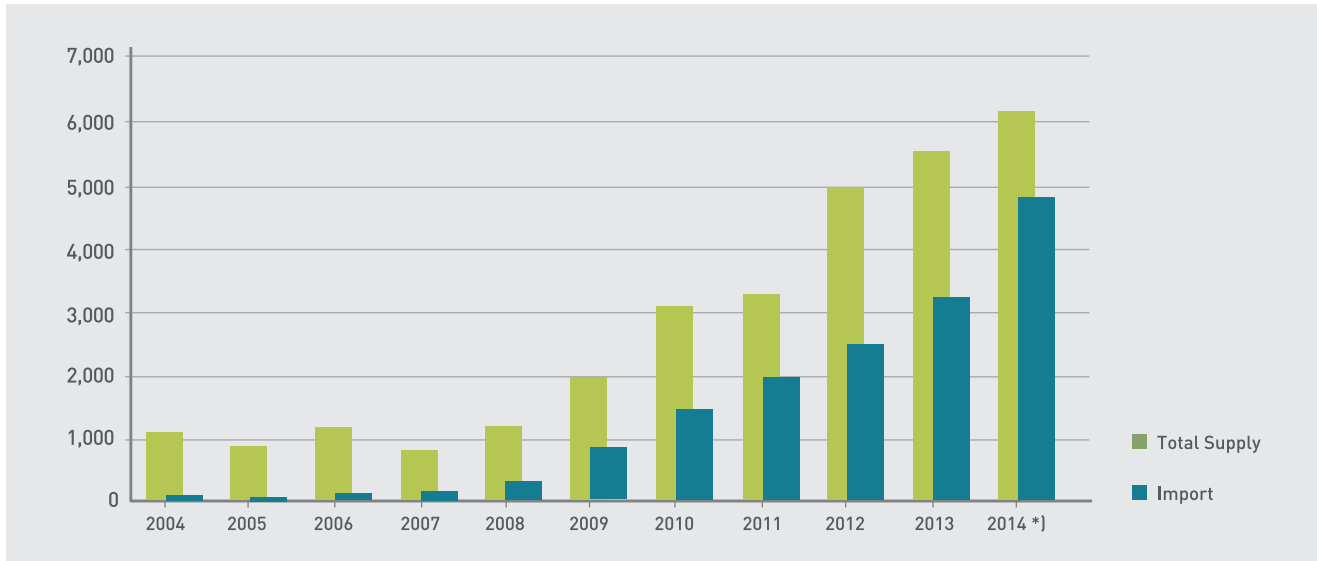
Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

Note :

- In 2012, share of fuel imports amounted to 54.25% [179.98 MBOE] of the total domestic fuel oil consumption
- The value of imports of RON 88 (premium) increased along with the increase in domestic demand
- Kerosene imports declined and abolished in 2009 due to LPG conversion policy

LPG IMPORTS

(Thousand Ton)



Source: Ministry of Energy and mineral Resources, reprocessed by NEC

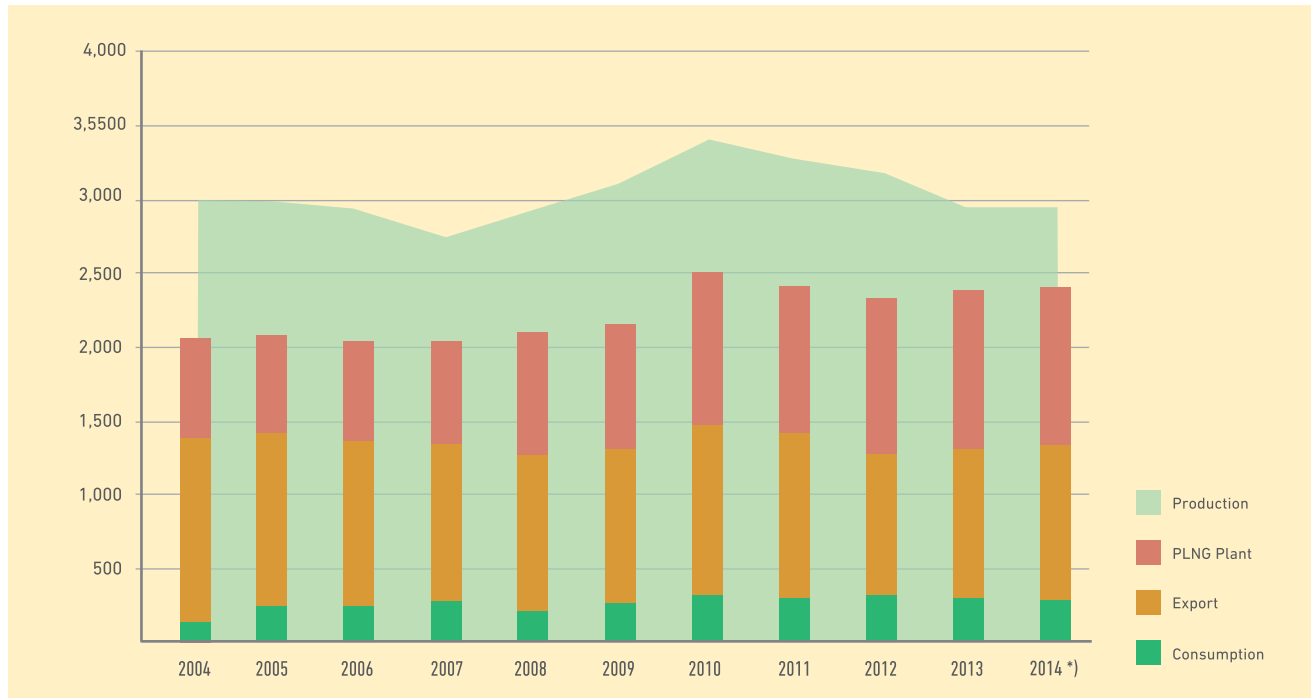
* Temporary Data

Note :

- In 2014, share of LPG imports amounted to 60.21% (4.9 million ton) of the total LPG supply
- The value of LPG supply increased along with the increasing the demand in residential sector due to LPG conversion policy

GAS SUPPLY

(Thousand MMSCF)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

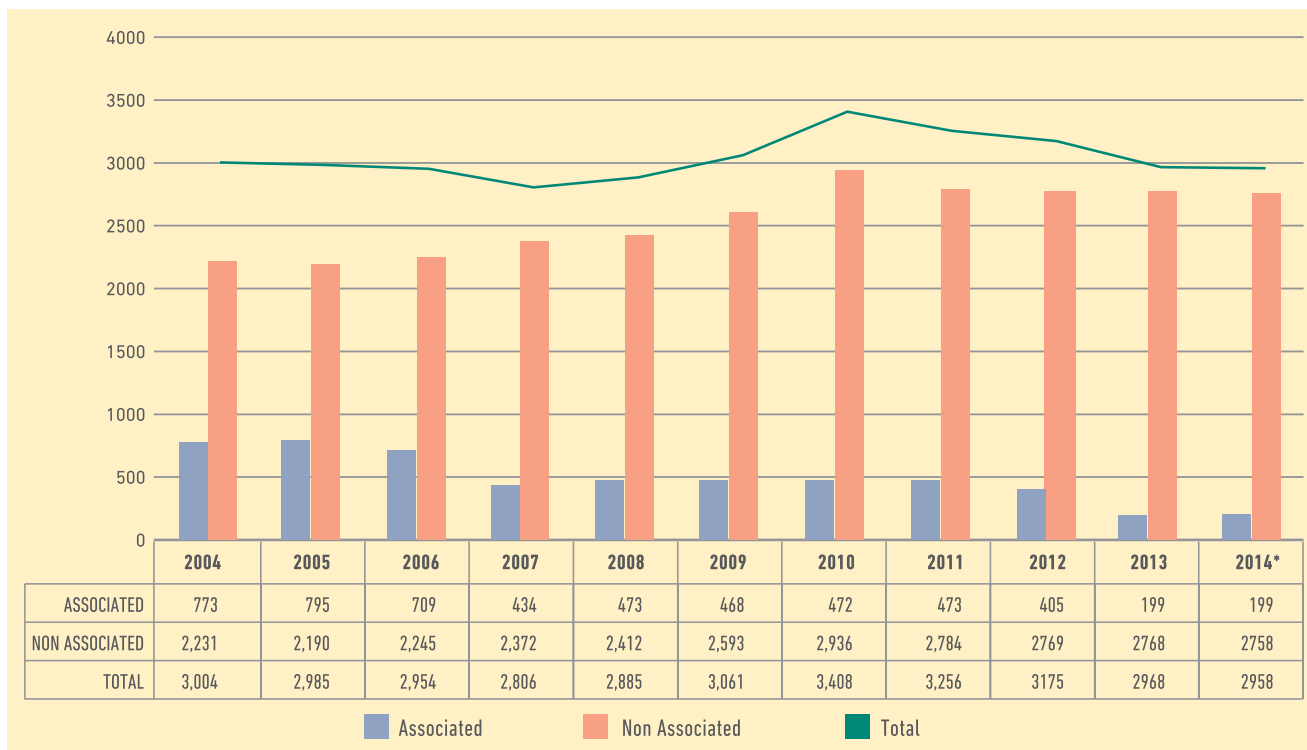
* Temporary data

Note :

- Products of LNG refinery allocated for exports
- Discrepancy between the production and the total usage of gas due to gas reinjection, own use and flare

GAS PRODUCTION

(Thousand MMSCF)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

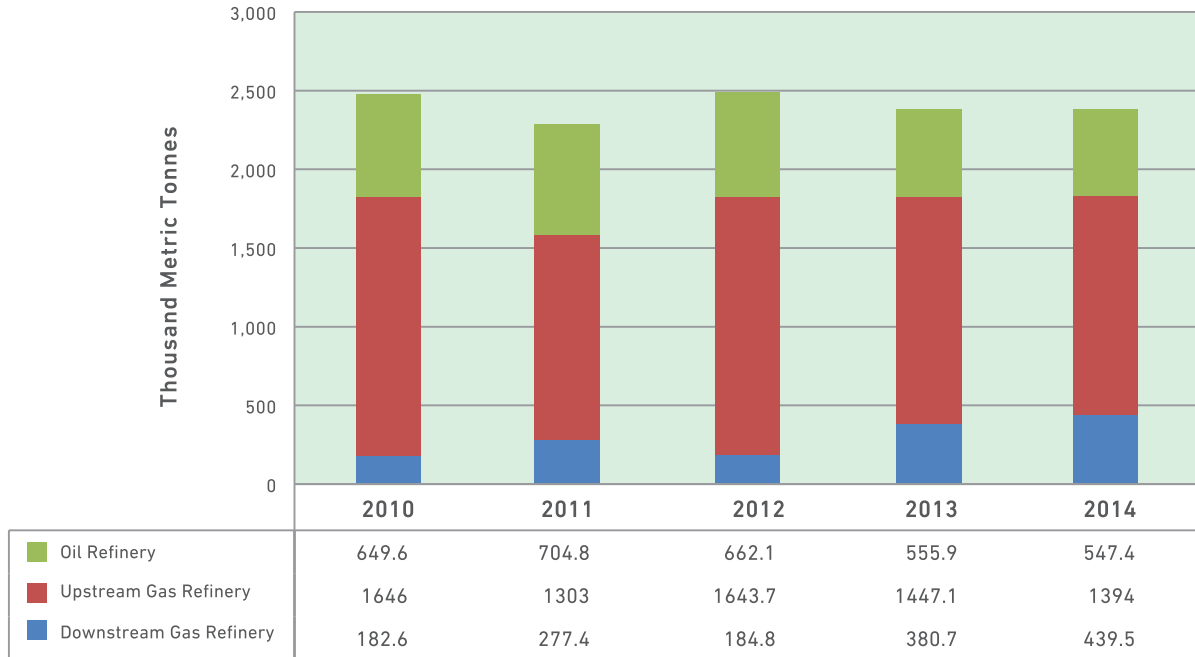
* Temporary data

Note :

Associated defined as a natural gas found in association with oil, either dissolved in the oil or as a cap of free gas above the oil.

LPG PRODUCTION

(Thousand Metric Tonnes)

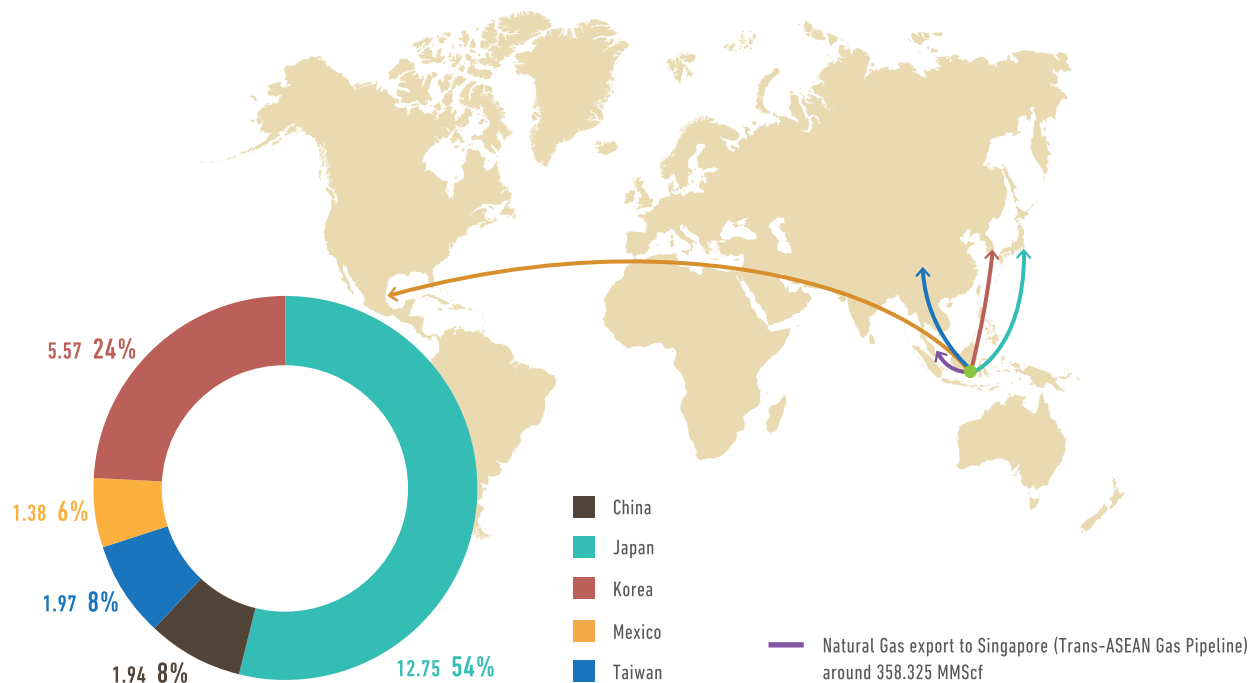


Source: Ministry of Energy and Mineral Resources 2015, reprocessed by NEC

- Upstream gas refinery means LPG production coming from refinery owned by PSC/KKKS company
- Downstream gas refinery means LPG production coming from processing gas company licensed by Government

GAS AND LNG EXPORTS 2014

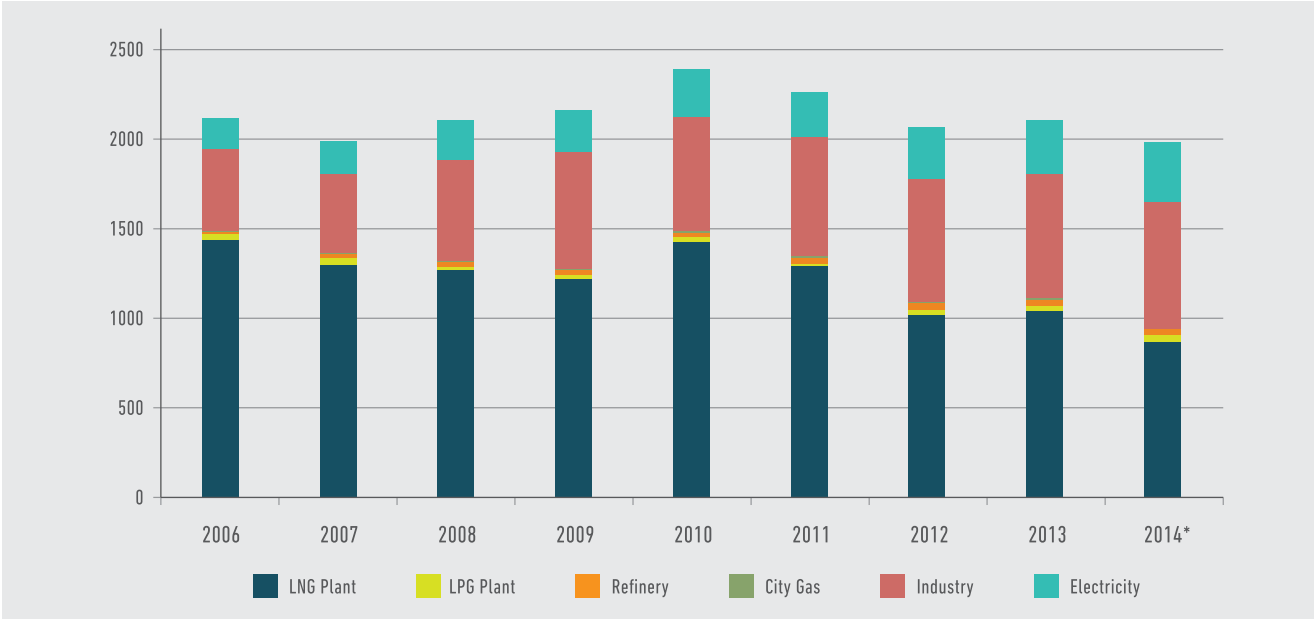
(Million Ton)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

NATURAL GAS CONSUMPTION

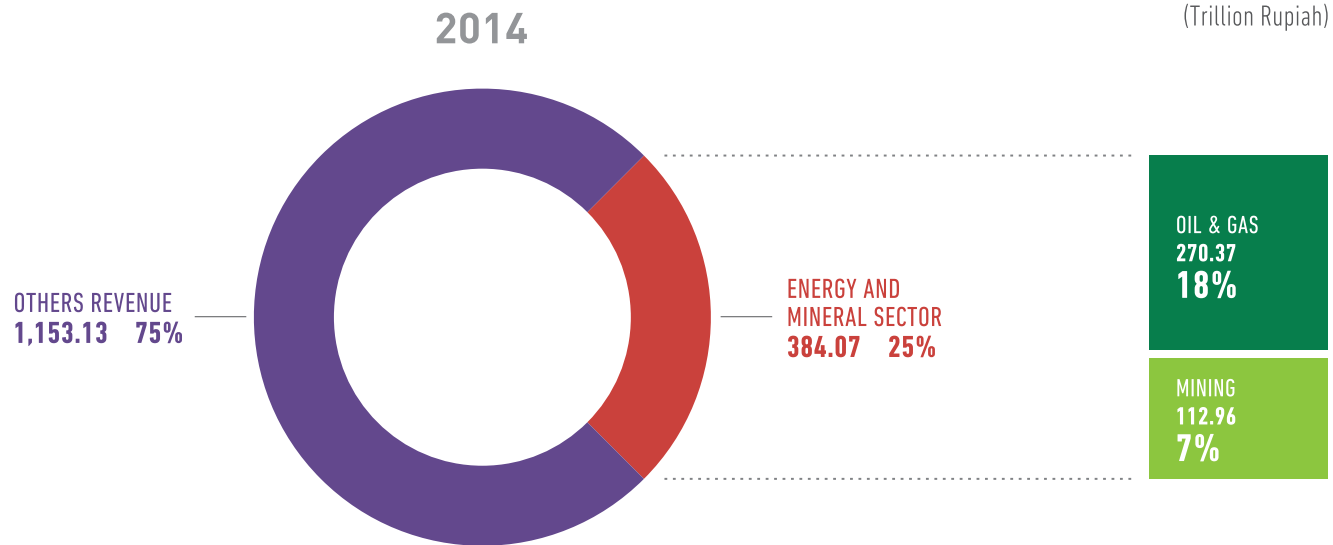
(Thousand MMSCF)



Source: Ministry of Energy and Mineral Resources 2015, reprocessed by NEC

* Unaudited data

STATE REVENUE FROM ENERGY AND MINERAL SECTOR

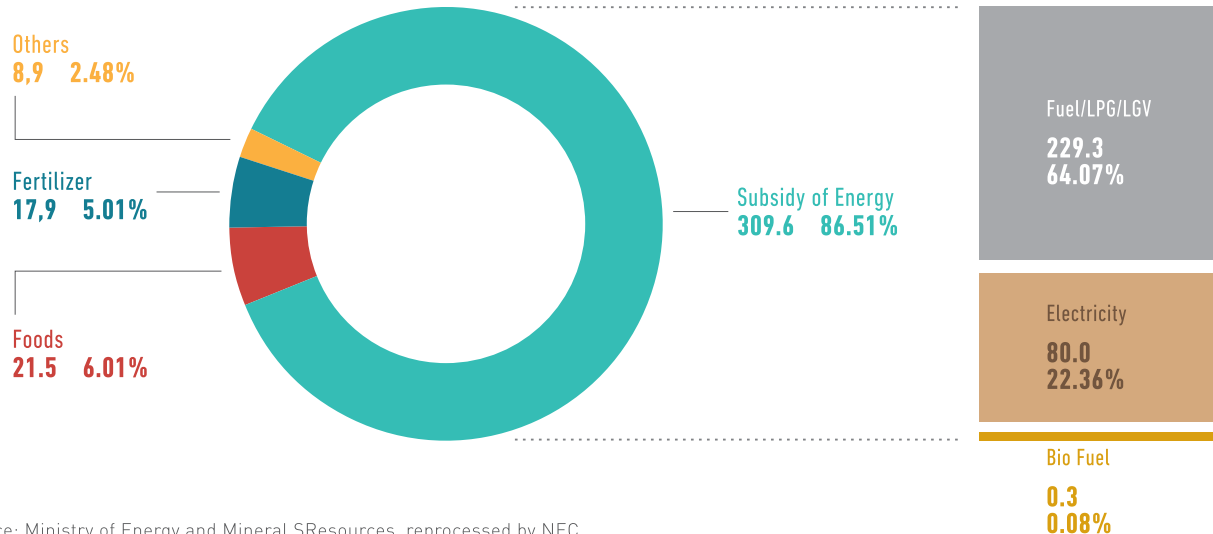


In 2014, the state revenue from energy and mining resources sector decreased by 14% when compared to the year 2013.

In 2013, the revenues reached 447 trillion IDR. This condition as a result of the export volume of oil and gas and mineral export restrictions policy and coal.

SHARES OF TOTAL ENERGY SUBSIDY, 2014

(Trillion Rupiah)

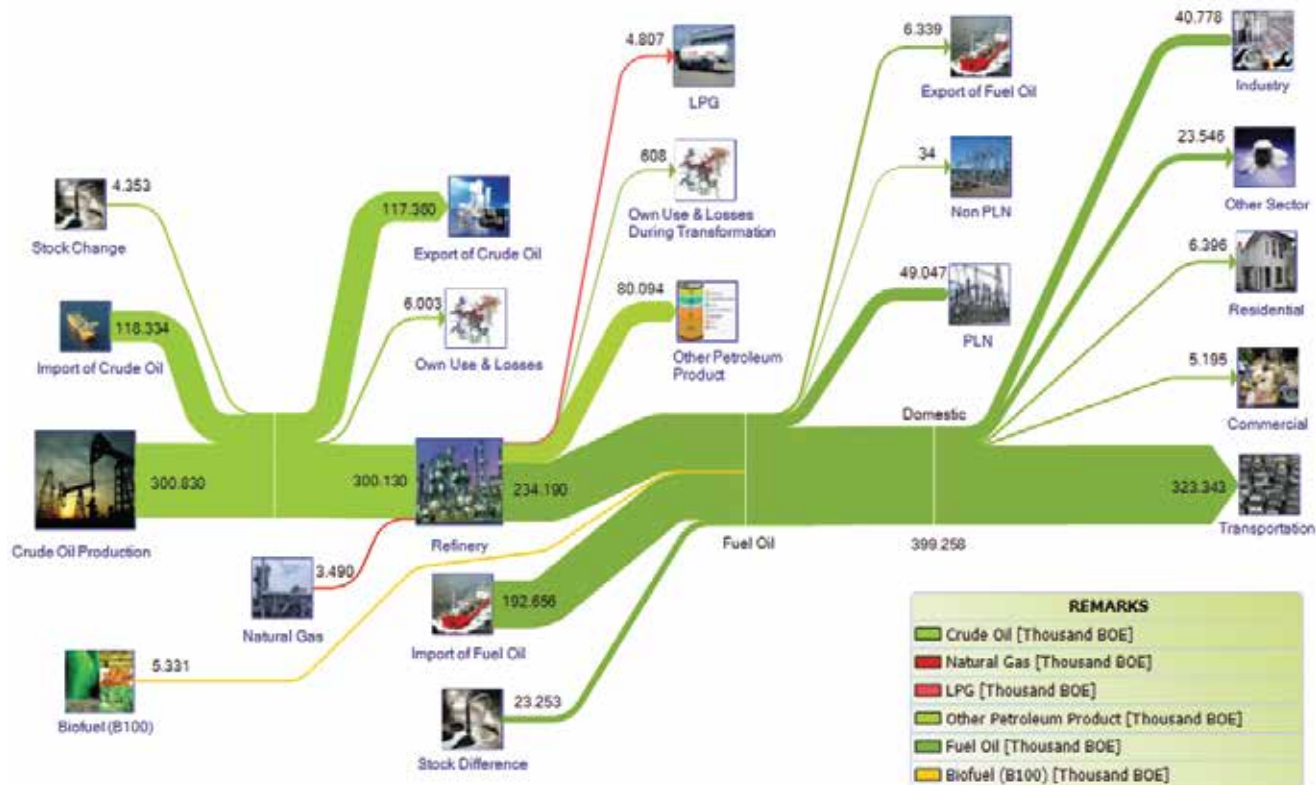


Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

* Temporary data in December 2014

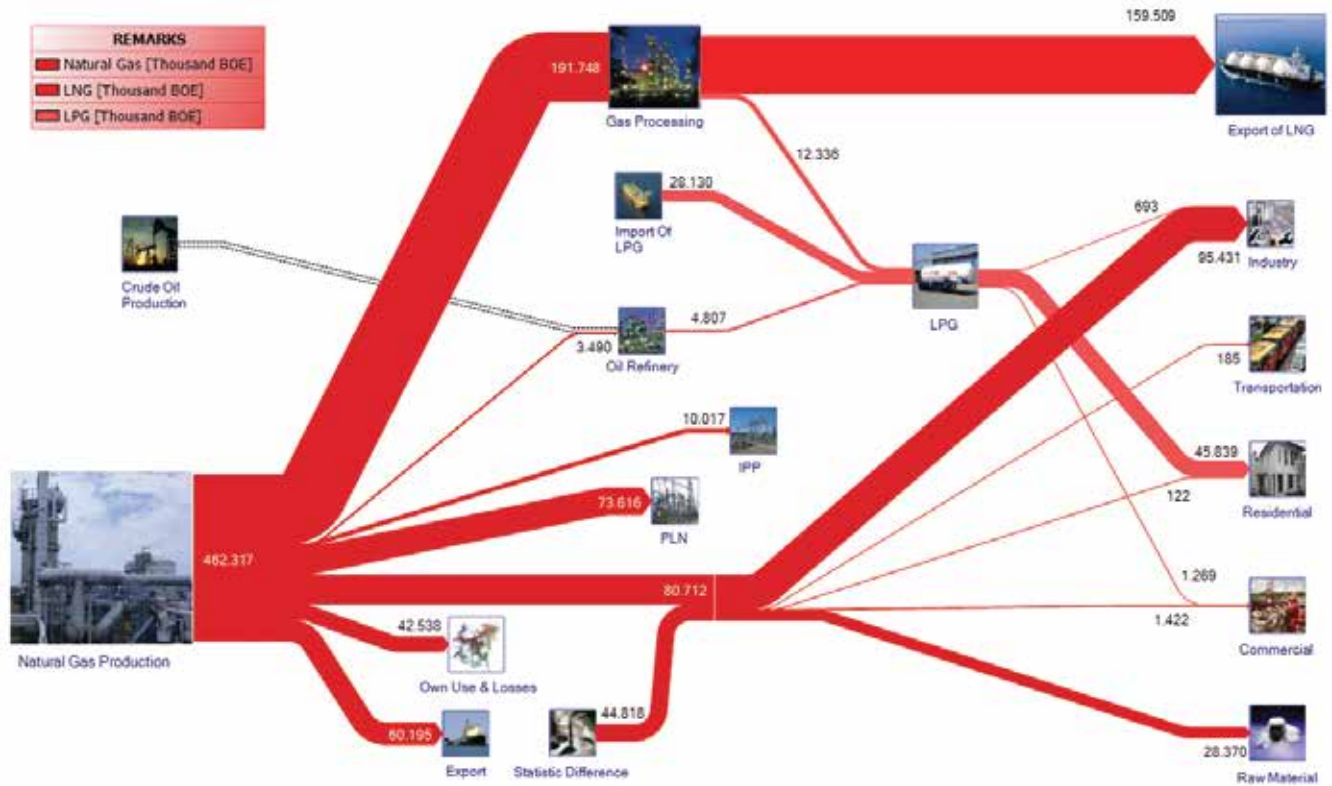
In 2014, energy subsidies increased by 3.3% compared to the year 2013 (299.8 trillion rupiah)

OIL FLOW, 2013

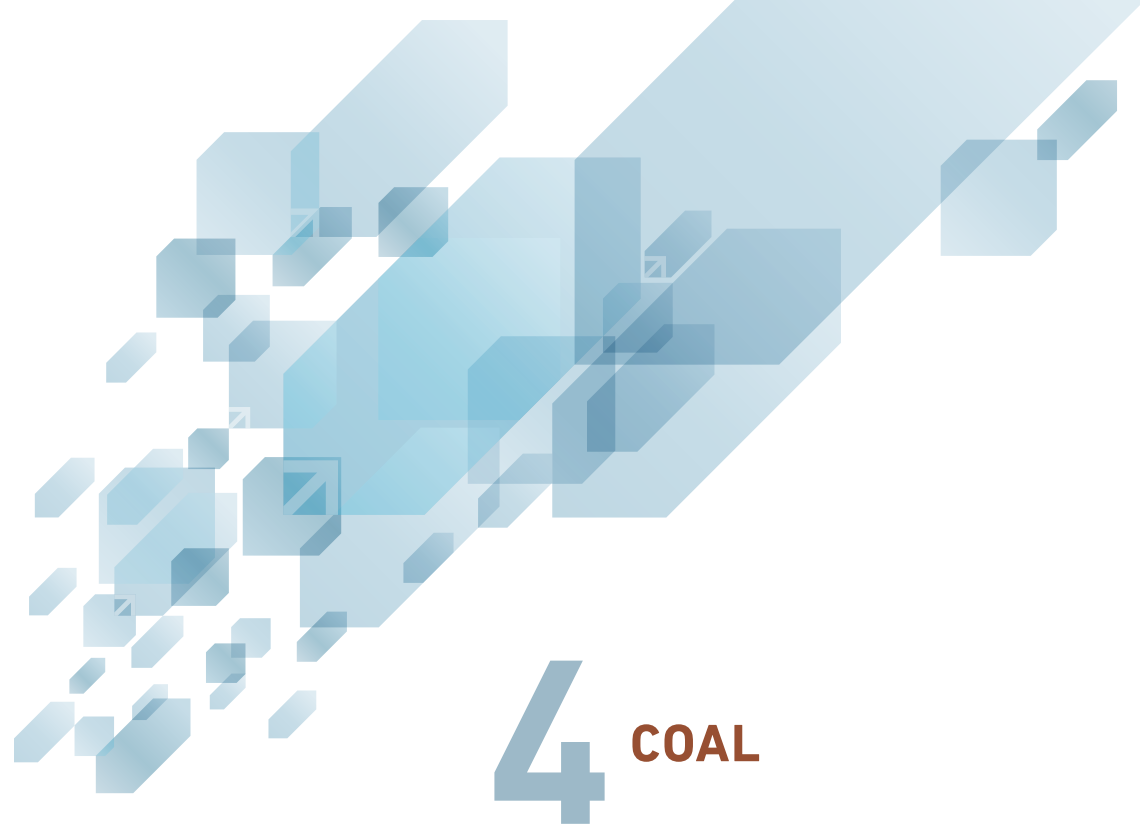


Source: National Energy Council, 2014

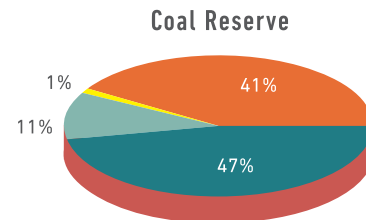
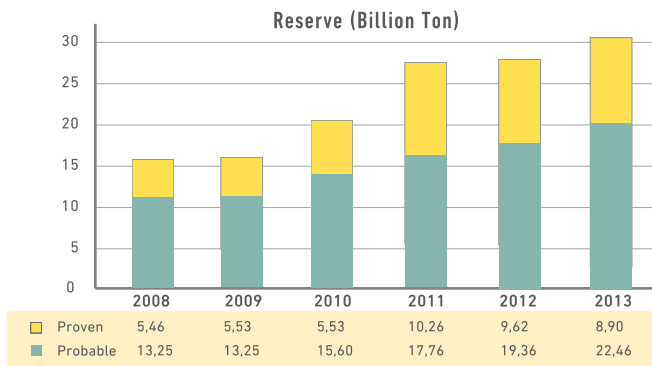
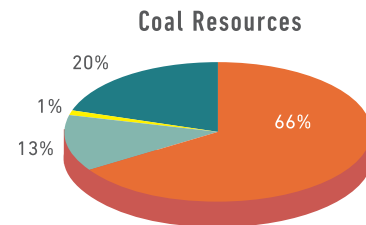
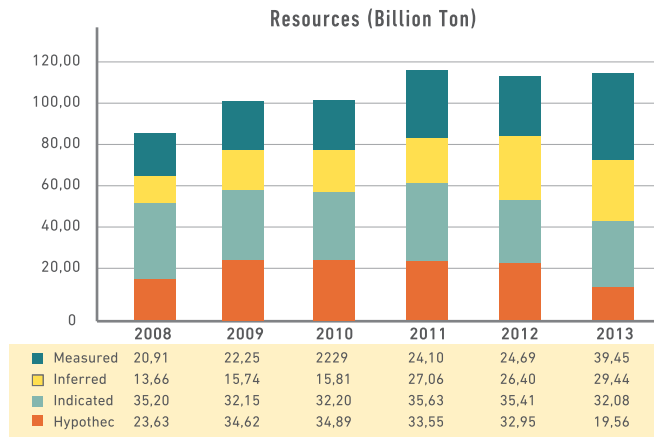
GAS FLOW, 2013



Source: National Energy Council, 2104

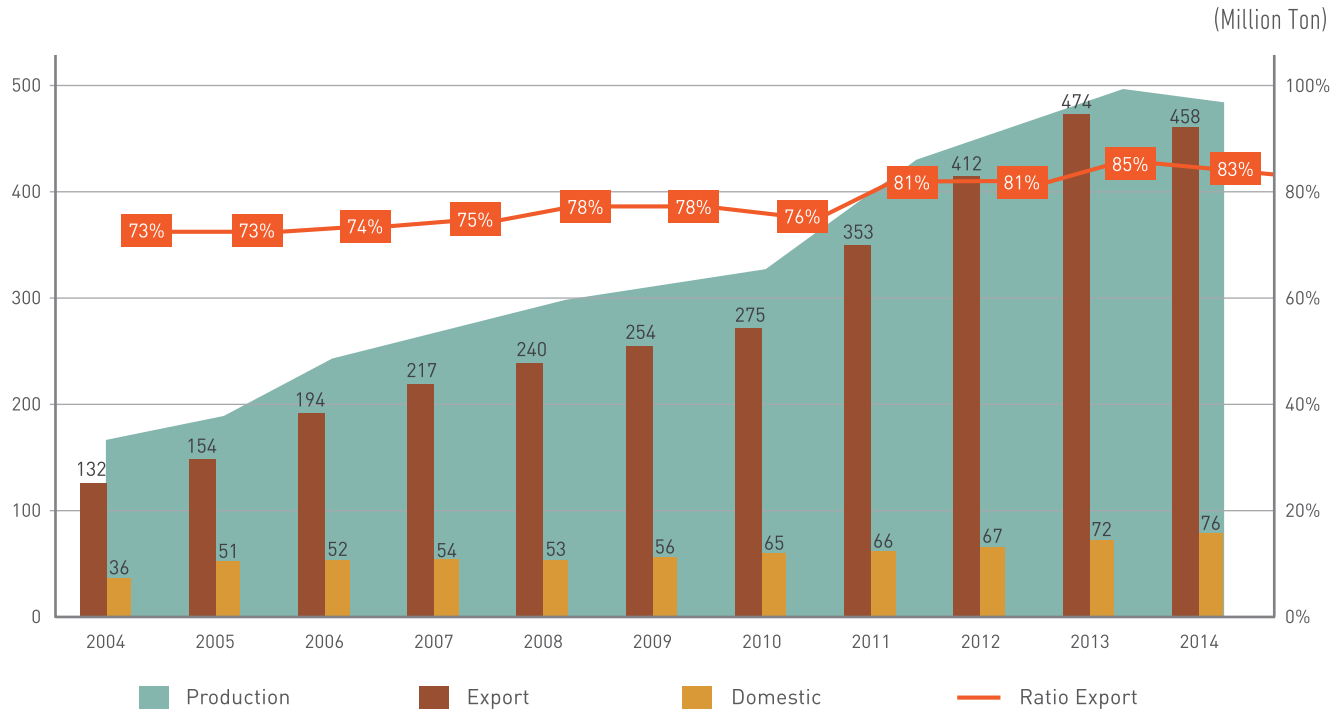


COAL RESOURCES AND RESERVE



The growth of total national coal resources and reserves are 5% and 11% per year.

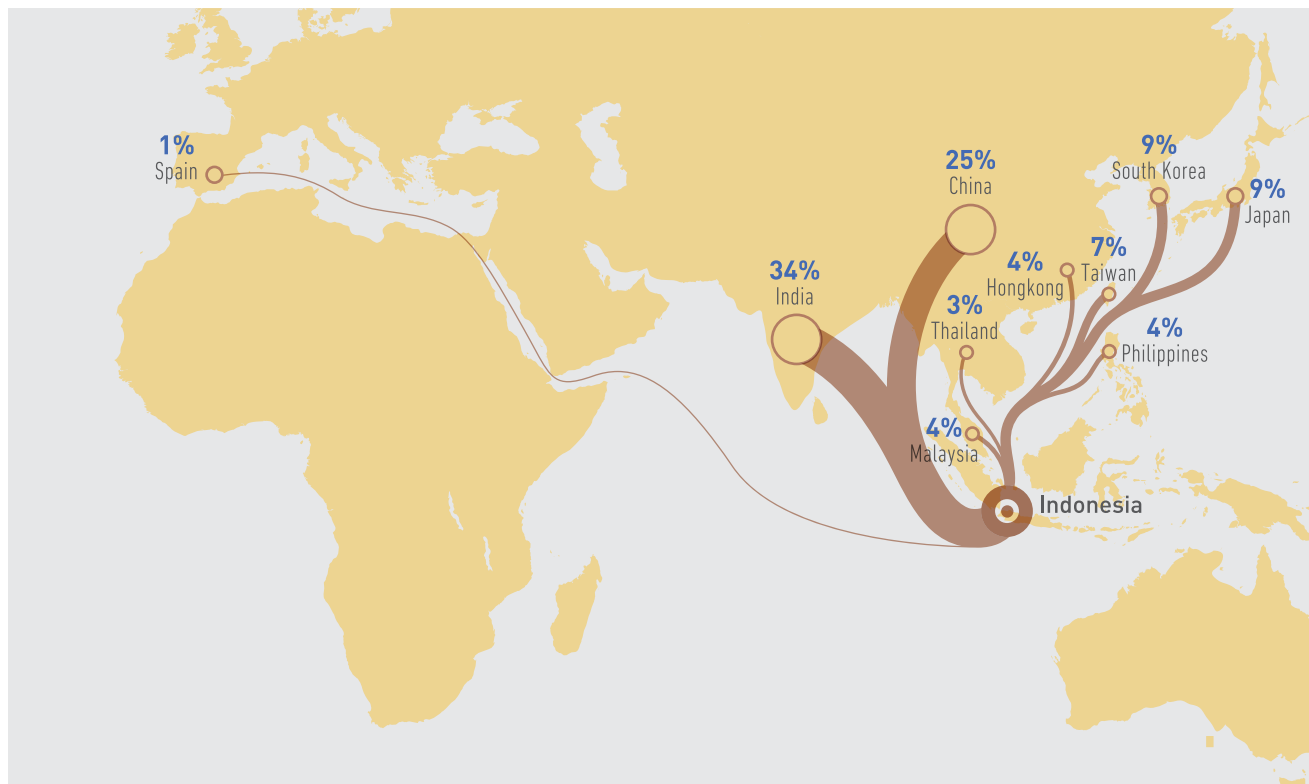
COAL UTILIZATION



Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC - *) April 2014

Coal production in 2014 decreased by 3.4% when compared to the previous year, however domestic consumption increased around 5.5% (76 million tonnes). This is due to start operating new power plants mostly from Fast Track Programme (FTP) projects.

COAL EXPORT BY COUNTRY DESTINATION

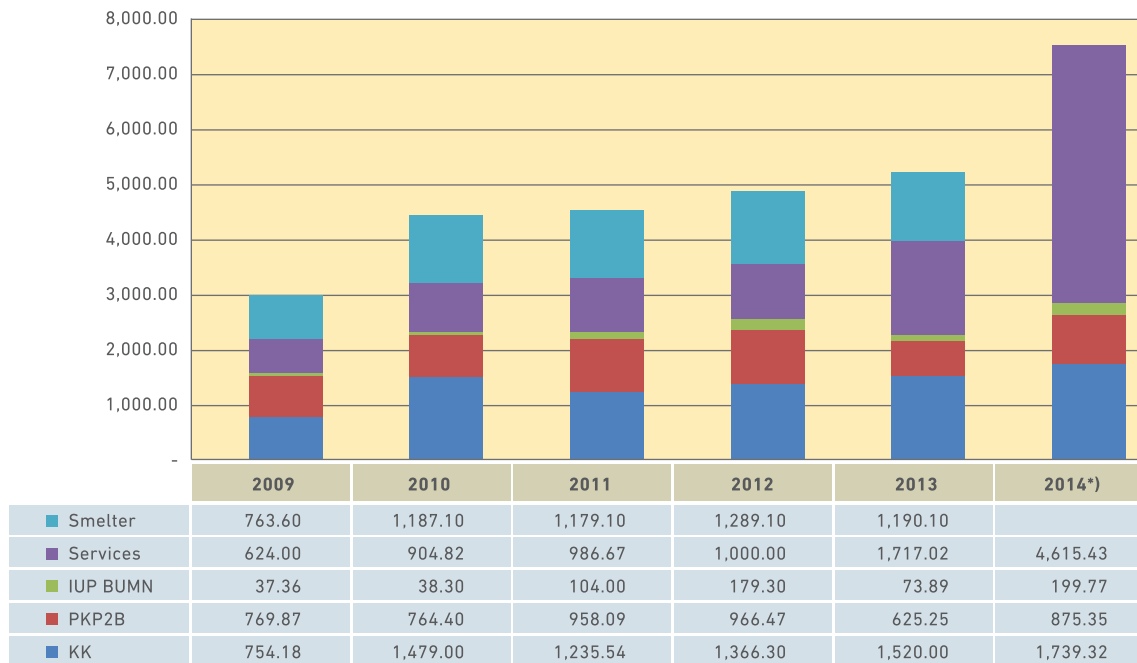


Source: Ministry of Energi and Mineral Resources, reprocessed by NEC

Note: Only from CCOW and PTBA

MINERAL AND COAL INVESTMENT

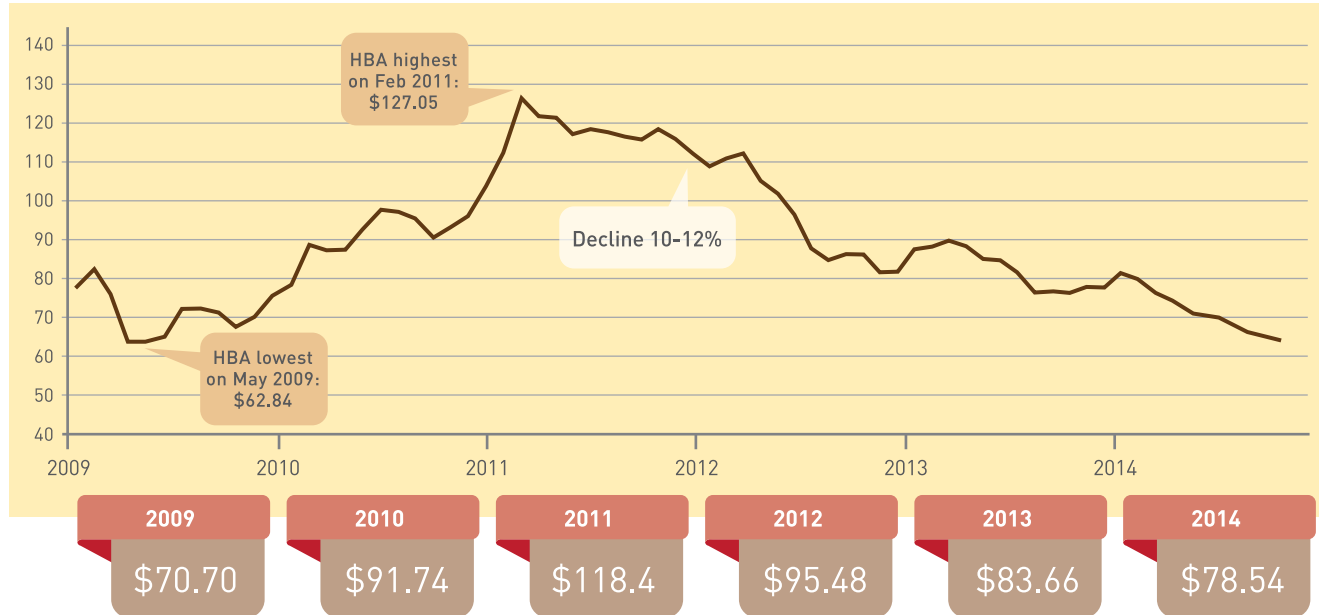
(Million US\$)



Source: Ministry of Energi and Mineral Resources, reprocessed by NEC

COAL REFERENCE PRICE (HBA)

(US Dollar)



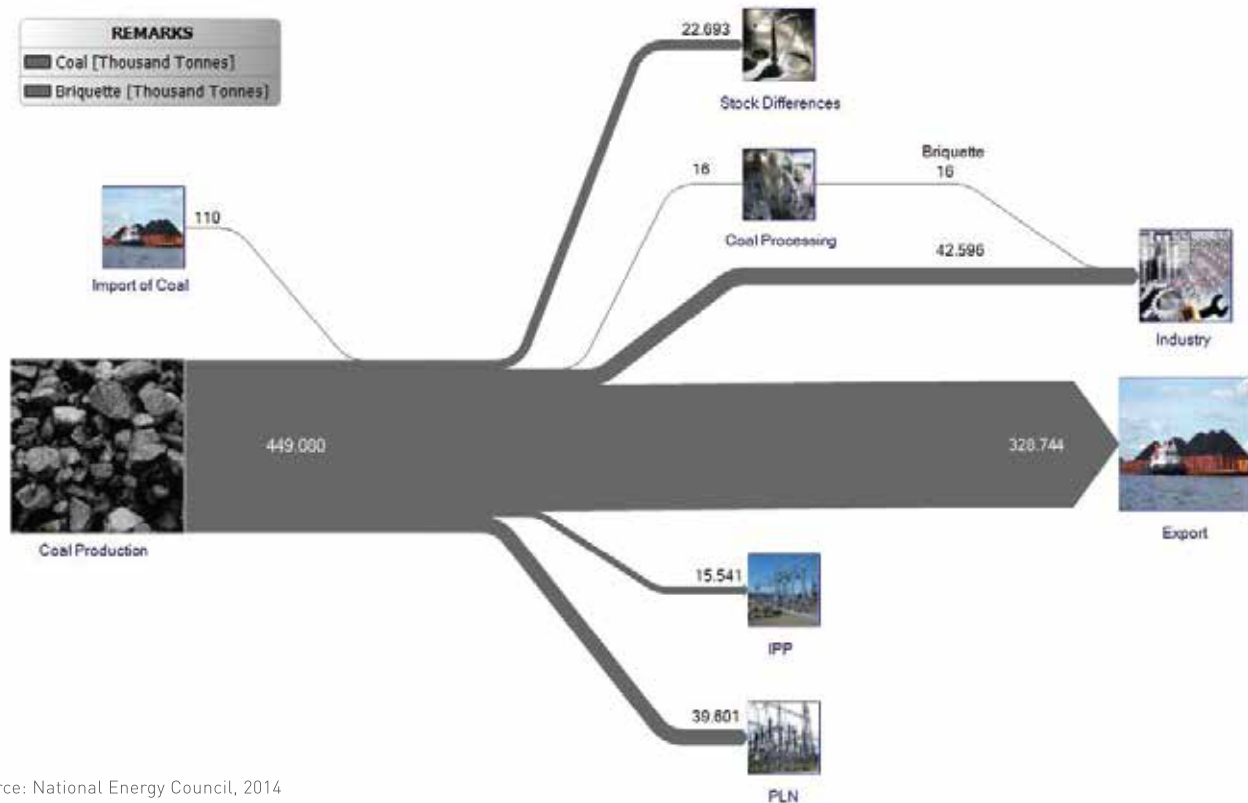
Source : Ministry of Energy and Mineral Resources,
reprocessed by NEC

Note : per January 2015

Decline in coal prices in the period 2012- 2013 caused by declining demand from some of the coal importing countries (China, India, and Japan) as well as the economic conditions from European countries are not yet stable.

The coal price increased in the period of 2011, due to supply disruptions from Australia, and increased demand from China and India.

COAL FLOW, 2013



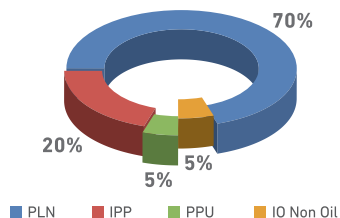
Source: National Energy Council, 2014



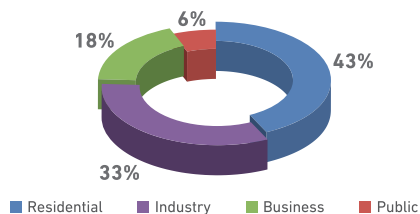
5 ELECTRICITY

THE OVERVIEW OF NATIONAL ELECTRICITY CONDITIONS, 2014

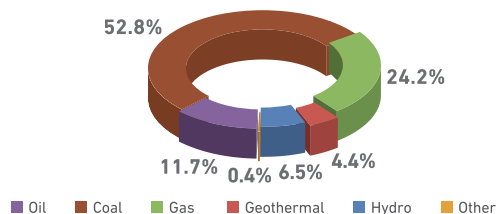
Percentage of Power Plant Capacity



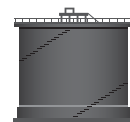
Percentage of Electricity Consumption By Sector



Percentage of Energy Mix

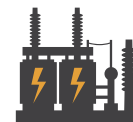


Share of fuel in Power Plant Energy Mix



11,65 %

Electricity Production (2014) *)



228 TWh

*) PLN and IPP



53.635 MW

Total Installed Capacity Power Plant (2014)

(PLN: 37.380 MW, IPP: 10.945 MW, PPU: 2.634 MW, IO Non BBM: 2.677 MW)

199 TWh



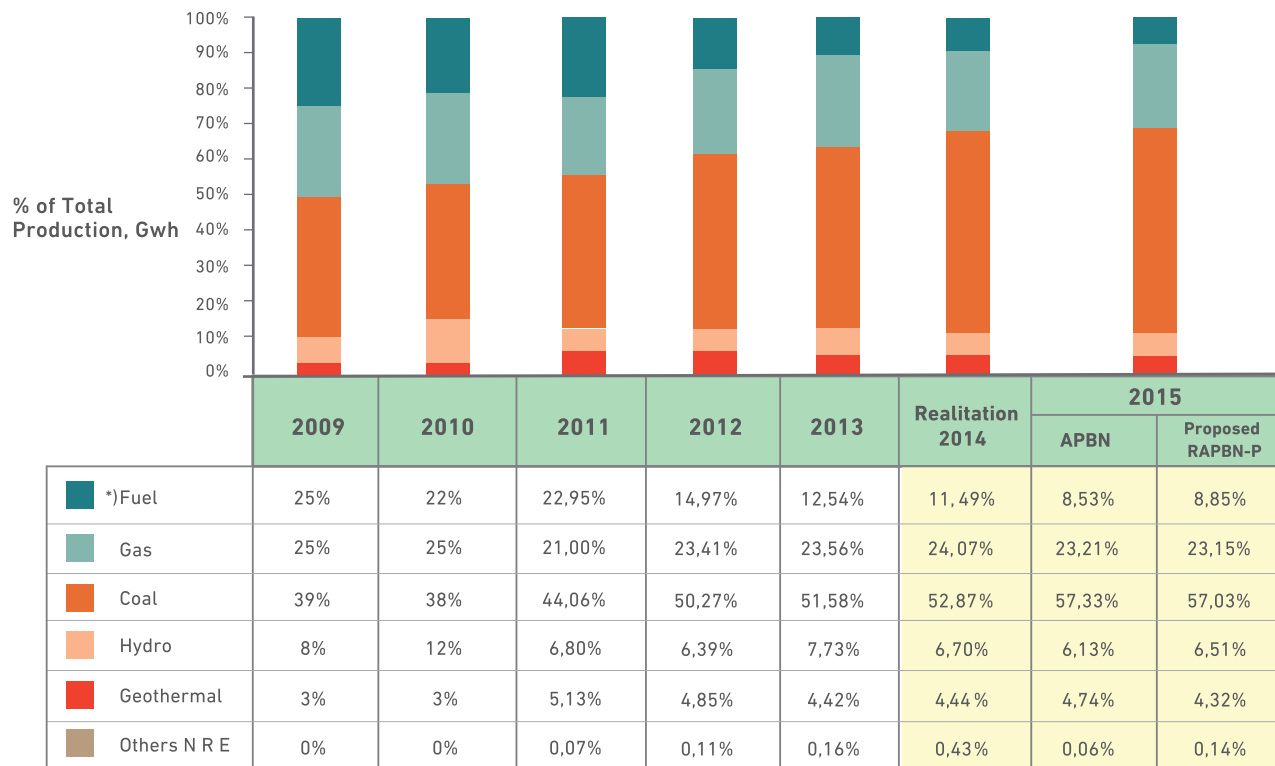
Electricity Consumption (2014)

84,35%



National Electrification Ratio (2014)

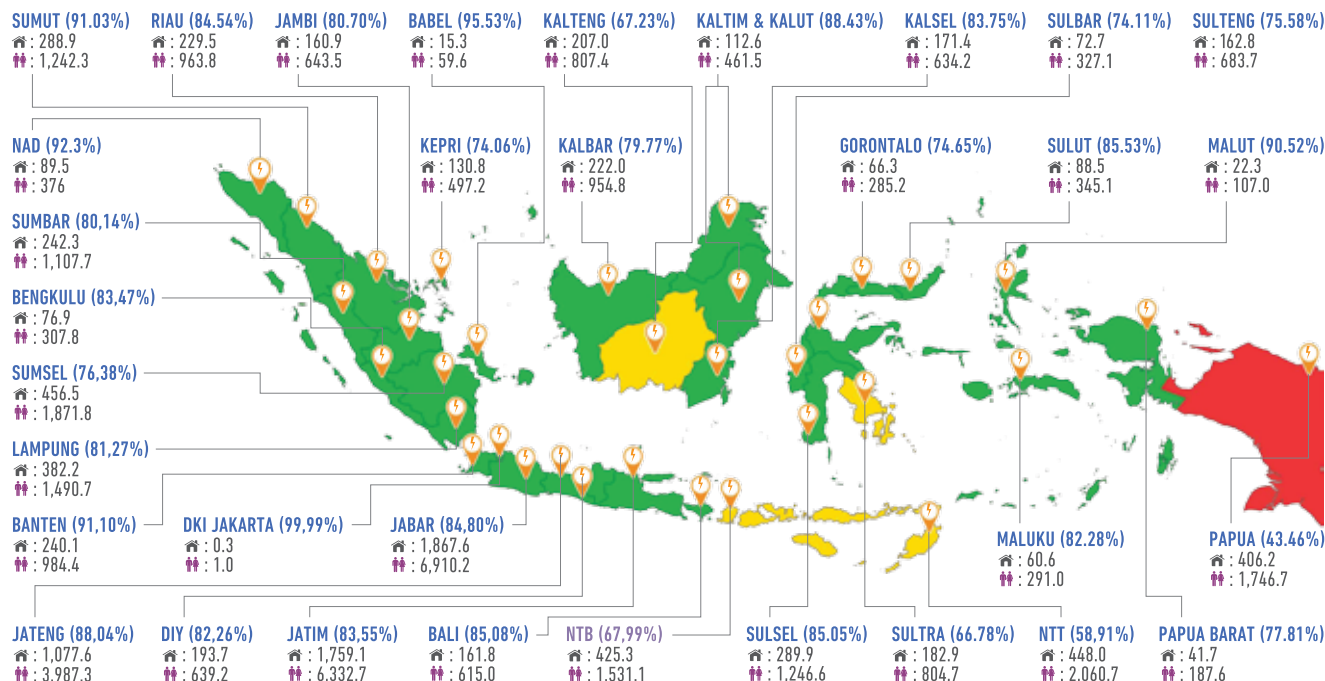
DEVELOPMENT AND ENERGY MIX TARGET IN POWER PLANT (2009 – 2015)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

*) Using Fuel includes Biodiesel 20%

ELECTRIFICATION RATIO 2014



Source: Ministry of Energy and Mineral Resources and BPS, March 2015, reprocessed by NEC

● > 70% ● > 50 - 70% ● < 50%

🏠 Non electrification (Thousand)

👤 Population not electric (Thousand People)

In 2014, electrification ratio amounted to 84.4, increased by 4.9% when compared to 2013 (80.5%). It is the result of more than 2 million households connected to electricity which has not been getting sufficient power connection.

TOTAL :

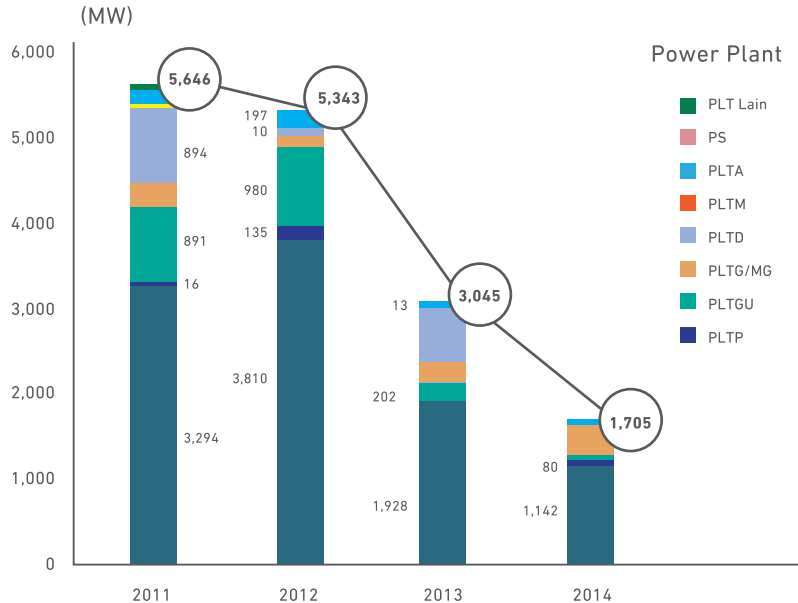
RE : 84.4%

🏠 : 10,146.82 Thousand Household

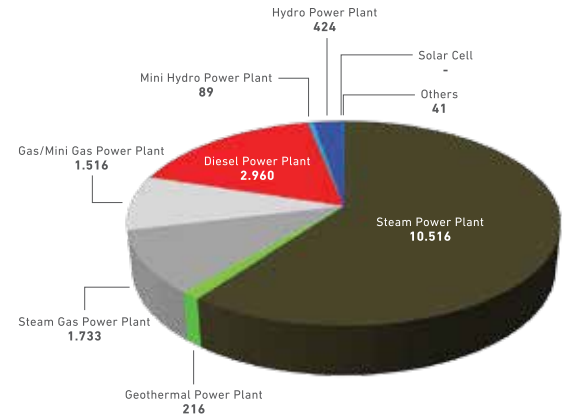
👤 : 39,572.58 Thousand People

PLN AND IPP POWER PLANT REALIZATION (2011 – 2014)

COD Realization by Year



Additional Capacity by Power Plant



Source : DJK dan RUPTL PLN 2015-2024

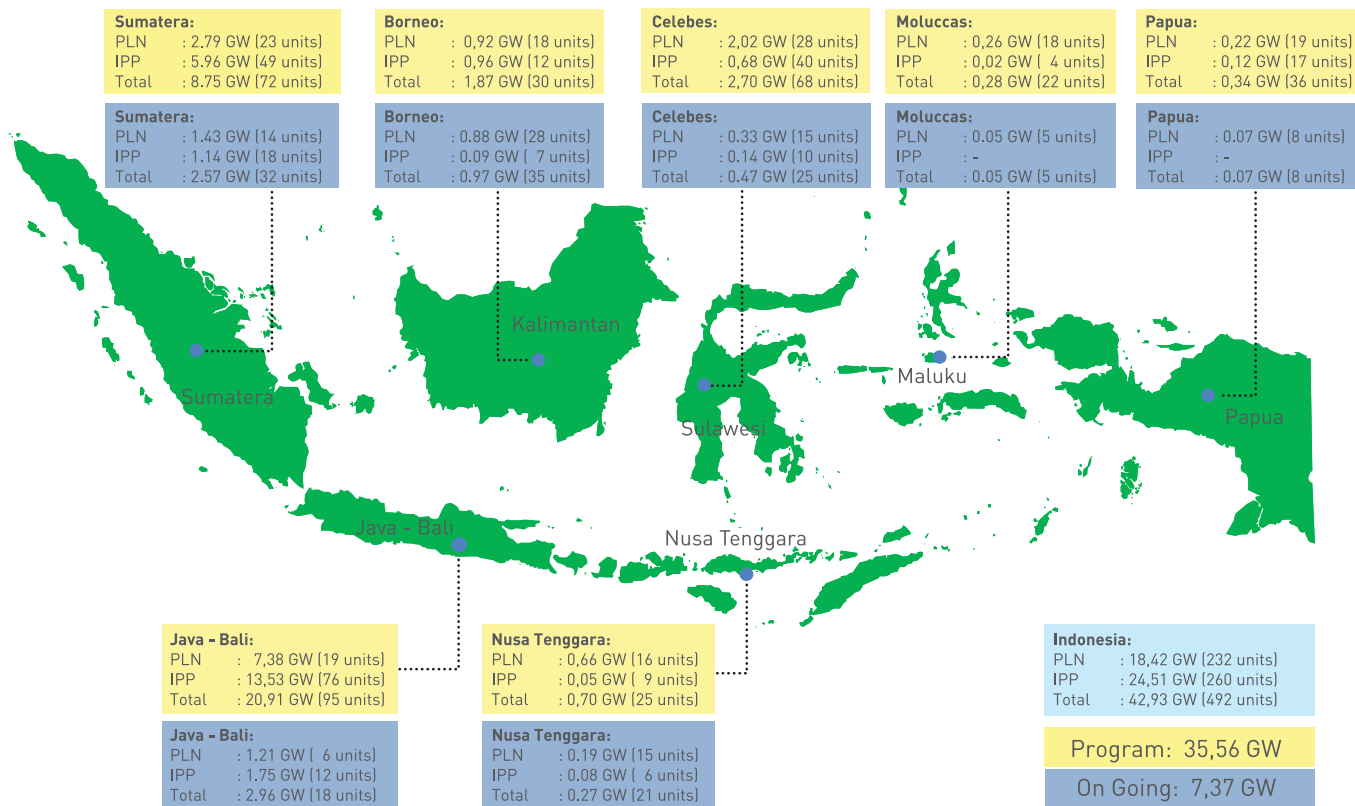
PLN & IPP Power Plant Capacity in the last 2009 increase from 30 GW to 48 GW in the last 2014



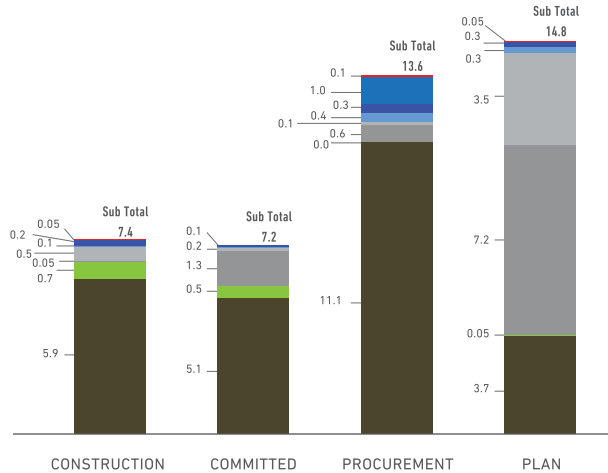
6

**PROGRAM OF POWER SUPPLY
INFRASTRUCTURE DEVELOPMENT**

POWER PLANT OF 35,5 GW AND 7,4 GW PROGRAM

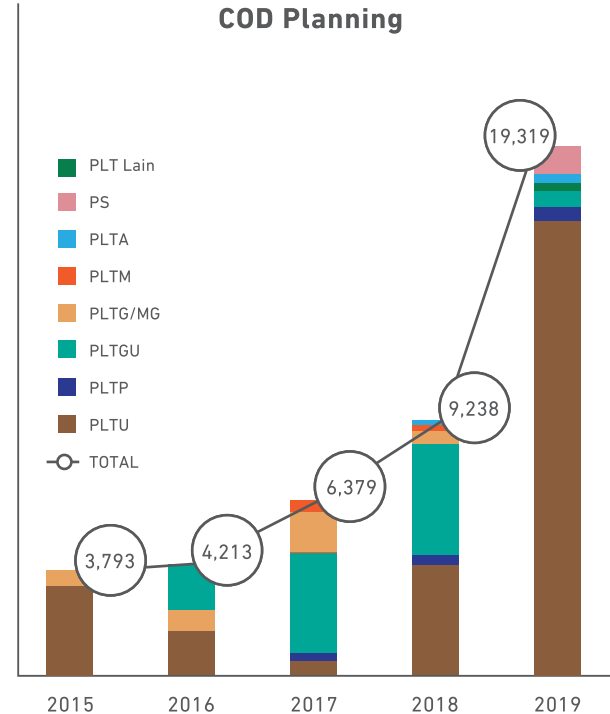


STATUS OF 35,5 GW + 7,4 GW POWER PLANT DEVELOPMENT (2015 - 2019)

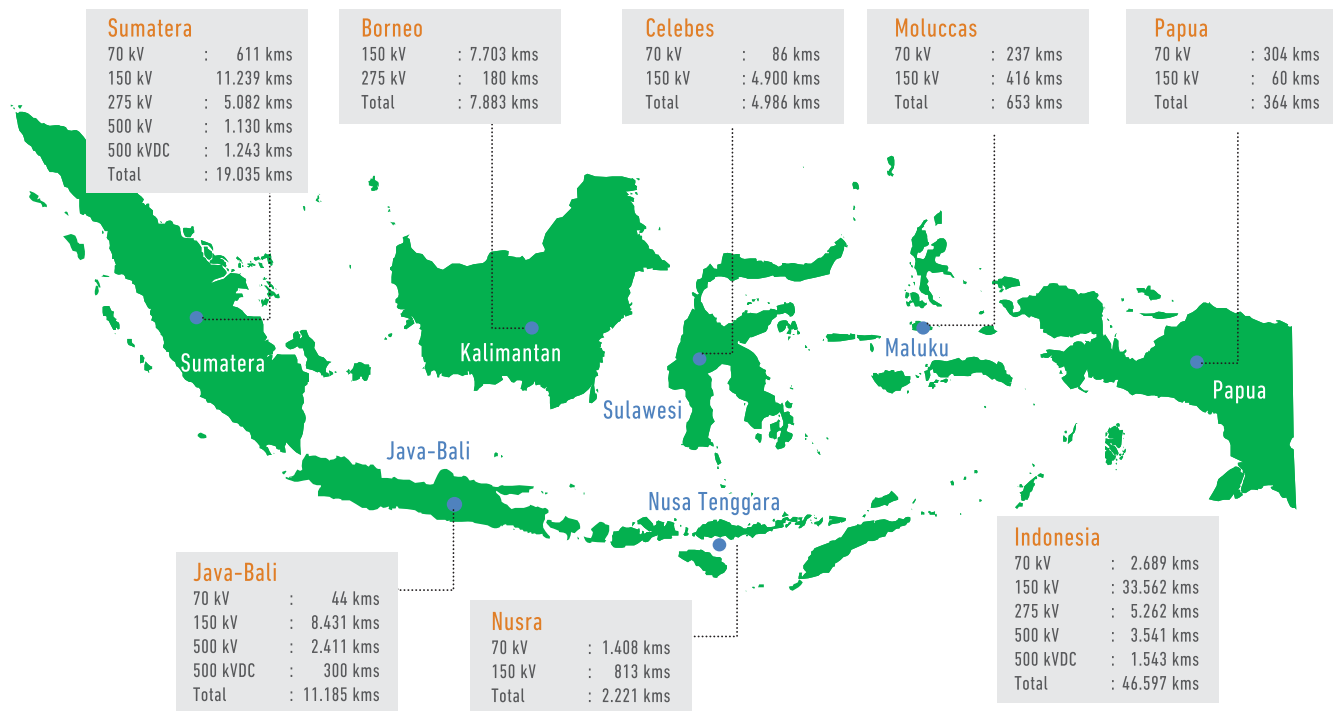


Description of Process

- Construction**
1. PLN : Contract with EPC had activated
 2. IPP : financial close (FC) had reached
- Procurement**
1. PLN : Process to contractor procurement EPC
 2. IPP : Process to procurement of PPA
- Committed**
1. PLN : Existing Budget
 2. IPP : PPA had signed and financial close (FC) had not reached
- Plan**
- In Preparation, Procurement and or Budgeting

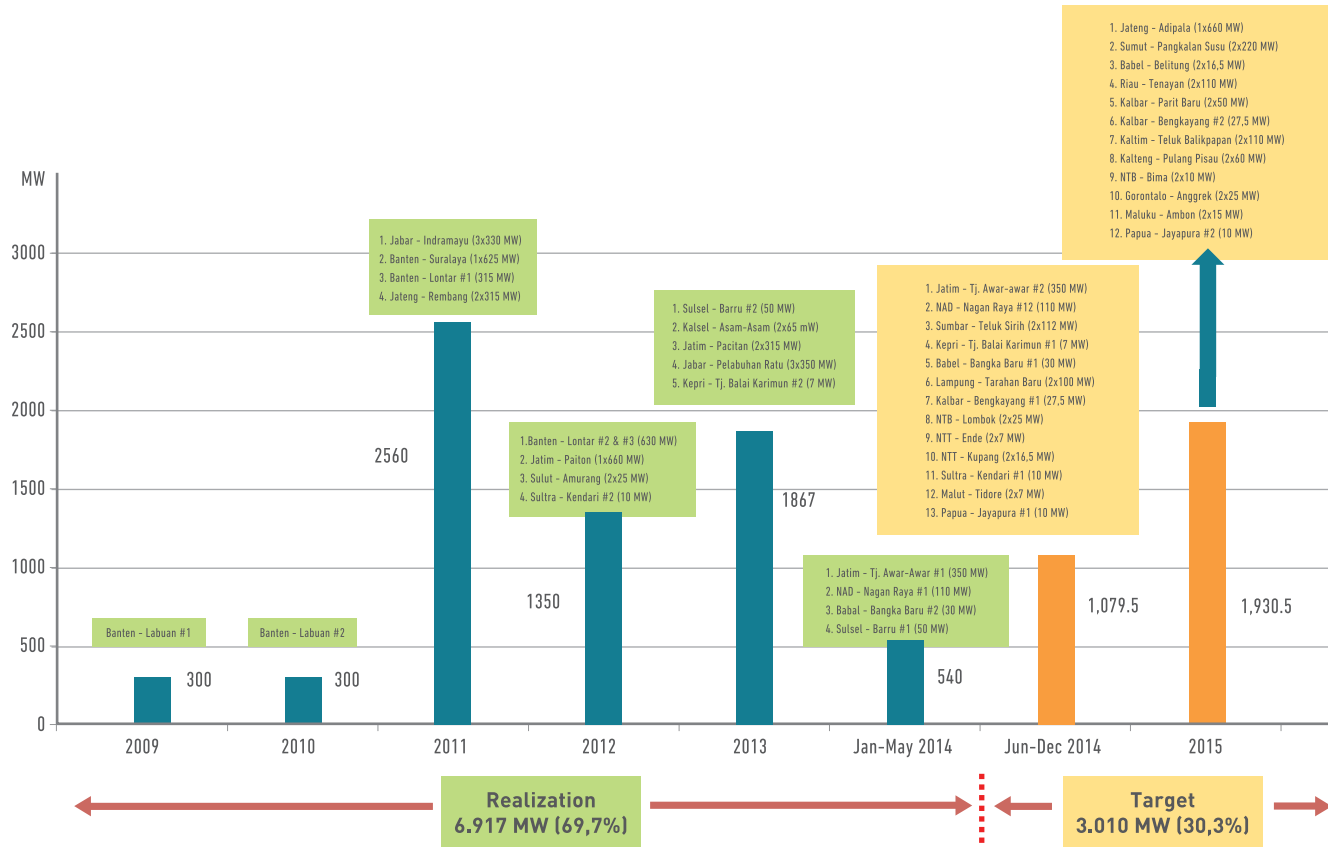


TRANSMISSION MAP OF 35,5 GW AND 7,4 GW PROGRAM



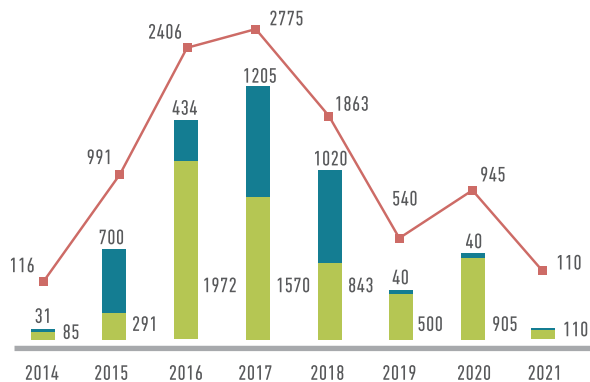
Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

REALIZATION AND COD TARGET OF POWER PLANT 10.000 MW FAST TRACK PROGRAM PHASE I

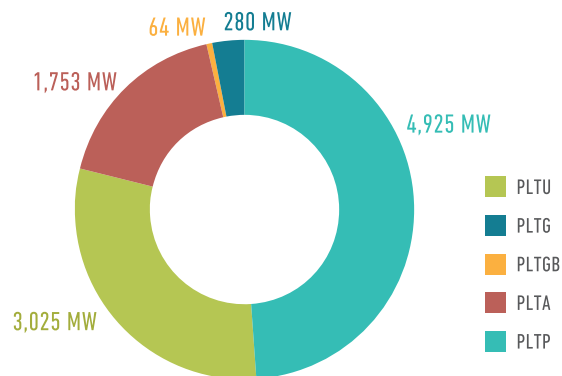


10,000 MW FAST TRACK PROGRAM PHASE II (FTP II)

COMMERCIAL OPERATION PLAN (MW)

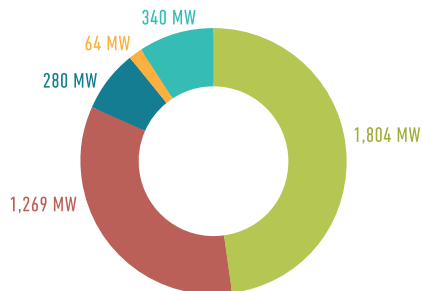


BY TYPE OF POWERPLANT (MW)

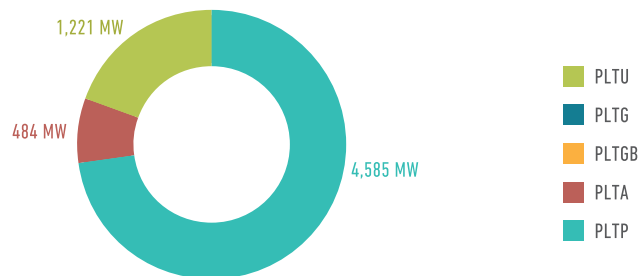


SHARE OF PROJECTS

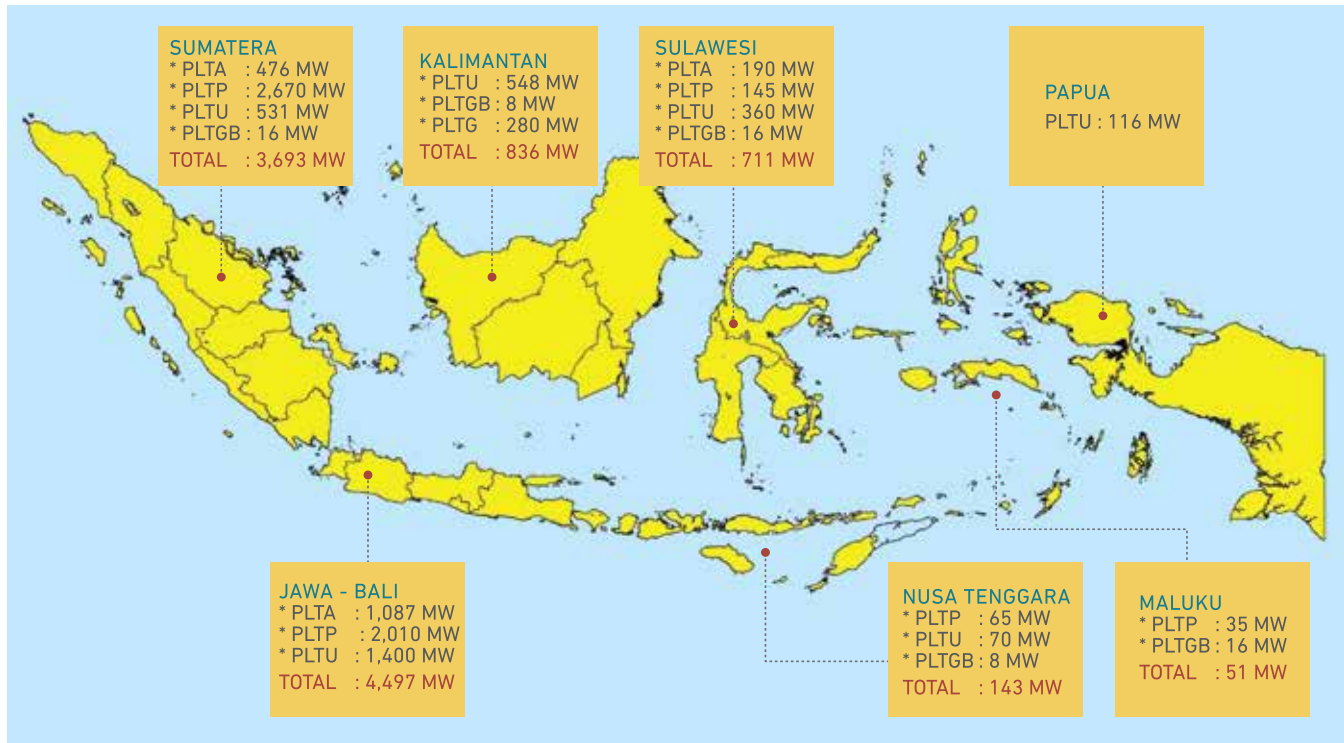
PLN (26 PROJECTS) : 3.757 MW (37%)



IPP (72 PROJECTS) : 6.290 MW (63%)



DISTRIBUTION OF 10,000 MW PROGRAM PHASE II



Source: Ministry of Energy and Mineral Resources 2012, reprocessed by NEC

*) EMR Ministerial Regulation Number 01 year 2012



7

**NEW AND RENEWABLE
ENERGY**

NEW AND RENEWABLE ENERGY RESOURCES

NEW AND RENEWABLE

NO.	NEW AND RENEWABLE ENERGY	RESOURCES	INSTALLED CAPACITY (IC)	RATIO RESOURCES/IC (%)
1	Hydro	75,000 MM	8.111,00 MW	10,81 %
2	Geothermal (MW)	29.475 MW	1.403,50 MW	4,8 %
3	Biomass	32.000 MW	1.740,40 MW	5,4%
4	Solar Cell	4,80 kWh / m2 / day	71,02 MW	-
5	Wind and Hybrid	3 - 6 m/s	3,07 MW	-
6	Ocean	61 GW ***)	0,01 MW ****)	-
7	Uranium	3.000 MW *)	30,00 MW **)	-

*) In Kalan – West Borneo

**) As Researches Center, non-energy

***) Source : Badan Litbang ESDM, 2014

****) Prototype BPPT

ENERGY SAVINGS POTENTIAL

NO.	SECTOR	ENERGY CONSUMPTION BY SECTOR YEAR OF 2012 (JUTA)	THE POTENTIAL OF ENERGY SAVINGS	TARGET OF POTENTIAL ENERGY SAVINGS (2025)
1	Industry	305 (39.7%)	10 - 30 %	17 %
2	Transportation	311 (40.4%)	15 - 35 %	20 %
3	Household	92 (12%)	15 - 30 %	15 %
4	Commercial	34 (4.4%)	10 - 30 %	15%
5	Others (Farming, Construction, dan Mining)	26 (3.4%)	25 %	-

*) Provisional figures up to December 2013 excluding biomass and non-energy use

NEW AND RENEWABLE POWER PLANT

Additional Power Plant Capacity from New and Renewable Energy in 2014 is 651 MW

CAPACITY INSTALLED CUMULATIVE UNTIL 2014 (MW)

NO. POWER PLANT	OFF GRID	ON GRID	TOTAL
1 GEOTHERMAL	-	1.403,50	1.403,50
2 HYDRO ENERGY	81,79	8.029,76	8.111,55
3 BIOENERGY	1.626,10	114,30	1.740,40
4 SOLAR ENERGY	66,02	5,00	71,02
5 WIND ENERGY	1,30	1,77	3,07
6 HYBRID	0,54	-	0,54
TOTAL	1.775,75	9.554,33	11.330,08

CAPACITY INSTALLED CUMULATIVE UNTIL 2013 (MW)

NO. POWER PLANT	OFF GRID	ON GRID	TOTAL
1 GEOTHERMAL	-	1.343,50	1.343,50
2 HYDRO ENERGY	70,3	7.503,30	7.573,60
3 BIOENERGY	1.626,00	90,50	1.716,50
4 SOLAR ENERGY	39,77	3,00	42,77
5 WIND ENERGY	-	1,33	1,33
6 HYBRID	0,54	-	0,54
TOTAL	1.736,61	8.941,63	10.678,24

GEOHERMAL POTENTIAL



P Geothermal Power Plant Existing

P Geothermal Power Plant Auction Preparation

P Geothermal Power Plant Plan

OPERATED GEOHERMAL POWER PLANTS : 1.403,5 MW

No.	Power Plants Geothermal	Total Capacity (MW)
1	Sibayak	12
2	Salak	377
3	Wayang Windu	227
4	Patuha	55
5	Kamojang	200
6	Darajat	270
7	Dieng	60
8	Lahendong	80
9	Ulubelu	110
10	Ulumbu	10
11	Mataloko	2,5

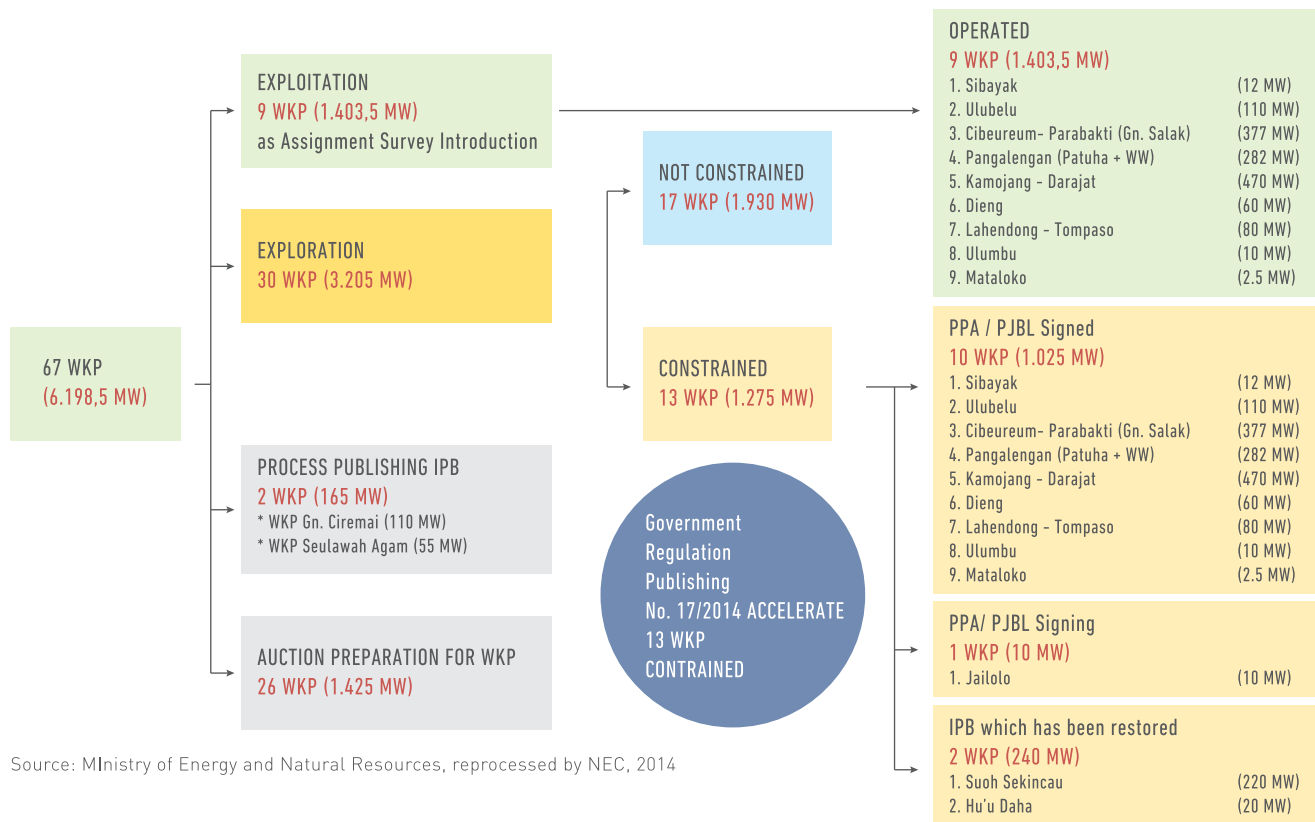
AUCTION PREPARATION OF GEOHERMAL POWER PLANTS : 1.425 MW

No.	Working Area	Capacity Plan (MW)	No.	Working Area	Capacity Plan (MW)	No.	Working Area	Capacity Plan (MW)
1	Bonjol	60	10	Songgoriti	20	19	Oka - Ile Ange	10
2	Gn. Talang - Bukit Kili	20	11	Gunung Lawu	165	20	Marana	20
3	Way Ratai	55	12	Sipoholon Ria-Ria	20	21	Bora Pulu	40
4	Gunung Endut	40	13	Kepahiang	110	22	Gunung Hamiding	2 x 5
5	Candi Umbul Telomoyo	55	14	Simbolon Samosir	110	23	Telaga Ranu	2 x 5
6	Gunung Wilis	20	15	Danau Ranau	110	24	Songa Wayaua	2 x 2,5
7	Gunung Arjuno Welirang	110	16	Graho Nyabu	110	25	Gn. Geureudong	2 x 55
8	Gunung Pandan	10	17	Suwawa	20	26	Gn. Galunggung	2 x 55
9	Gunung Gede Pangrango	55	18	Sembalun	20			

PLAN : 3.370 MW

No.	Working Area	Develop ment Plan (MW)	COD Plan
1	Sibual - Buali	3 x 110	2016 2017 2018
2	Hutulais - Tambang Sawah	2 x 55	2018 2019
3	Lumut Balai	4 x 55	2016 2018 2019
4	Sungai Penuh	2 x 55	2019
5	Karaha - Cakrabuana	1 x 30 2 x 55	2016 2019
6	Buyan Bratan (Bedugul)	1 x 10	2018
7	Kotamobagu	2 x 20	2019
8	Iyang - Argopuro	1 x 55	2019
9	Tulehu	20	2018
10	Cibuni	10	2017
11	Ciater	30	2019
12	Liki Pinangawan Muaralaboh	220	2017 2018
13	Gn. Rajabasa	220	2018 2019
14	Jaboi	10	2018
15	Sorik Marapi - Roburan Sampura	240	2018 2019
16	Cisolok Cisukarame	45	2017
17	Gn. Tangkuban Perahu	110	2017
18	Gn. Tampomas	40	2018
19	Gn. Ungaran	55	2017
20	Sokoria	6 x 5	2017 2018 2019
21	Atadei	10	2016
22	Jailoto	10	2019
23	Suoh Sekincau	220	2018 2019
24	Hu'u Daha	20	2018 2019
25	Kaldera Danau Banten	110	2019
26	Rantau Dedap	220	2018
27	Blawan - Ijen	130	2019
28	Telaga Ngebel	165	2019
29	Baturaden	220	2018 2021
30	Guci	55	2019
31	Seulawah Agam	55	2020
32	Ciremai	110	2020

STATUS OF GEOTHERMAL WORKING AREA (WKP)

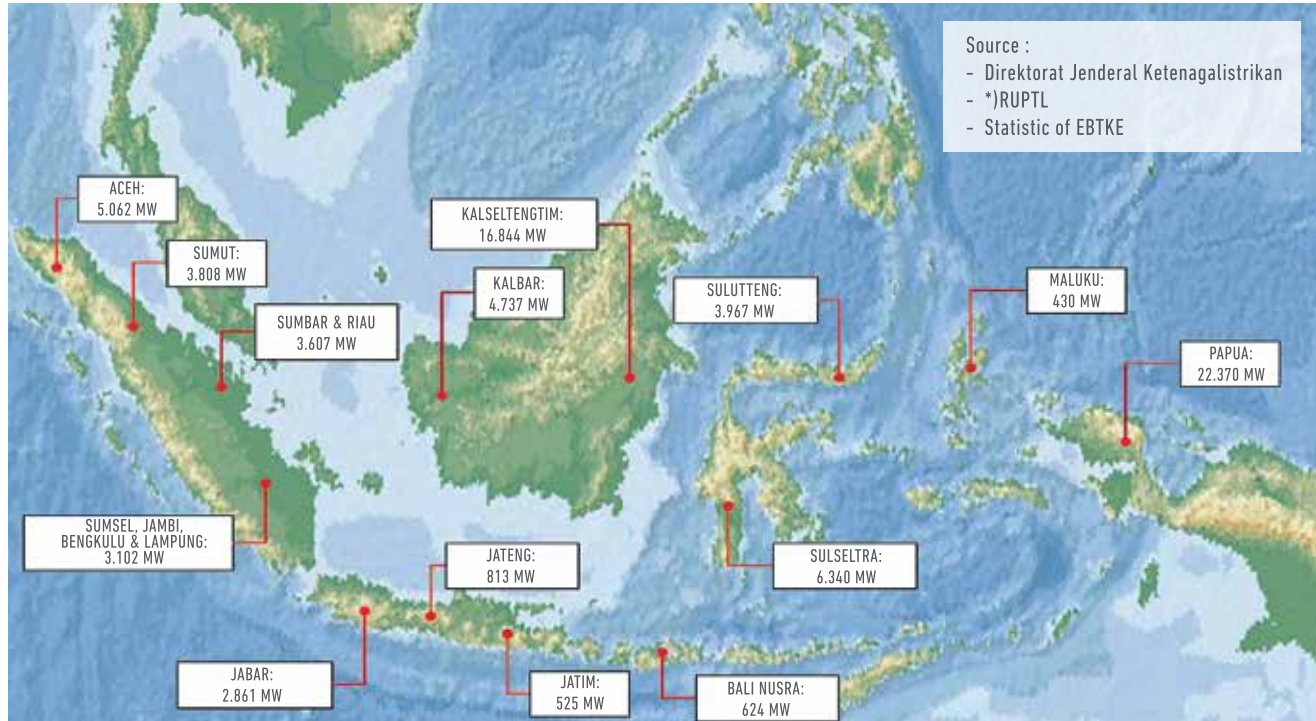


Source: Ministry of Energy and Natural Resources, reprocessed by NEC, 2014

Note:

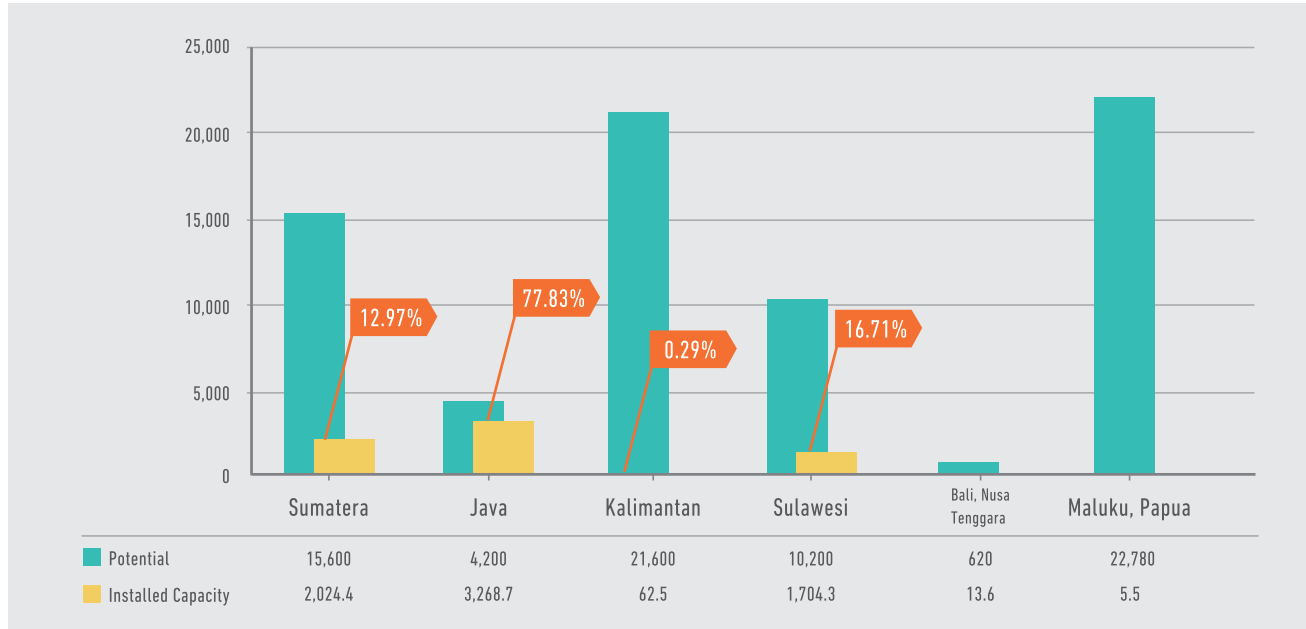
Currently there are 13 enterprises are doing an Assignment Survey Introduction and results are expected can be set to be WKP

HYDRO POTENTIAL MAP



HYDRO POTENTIAL AND INSTALLED CAPACITY

(MW)



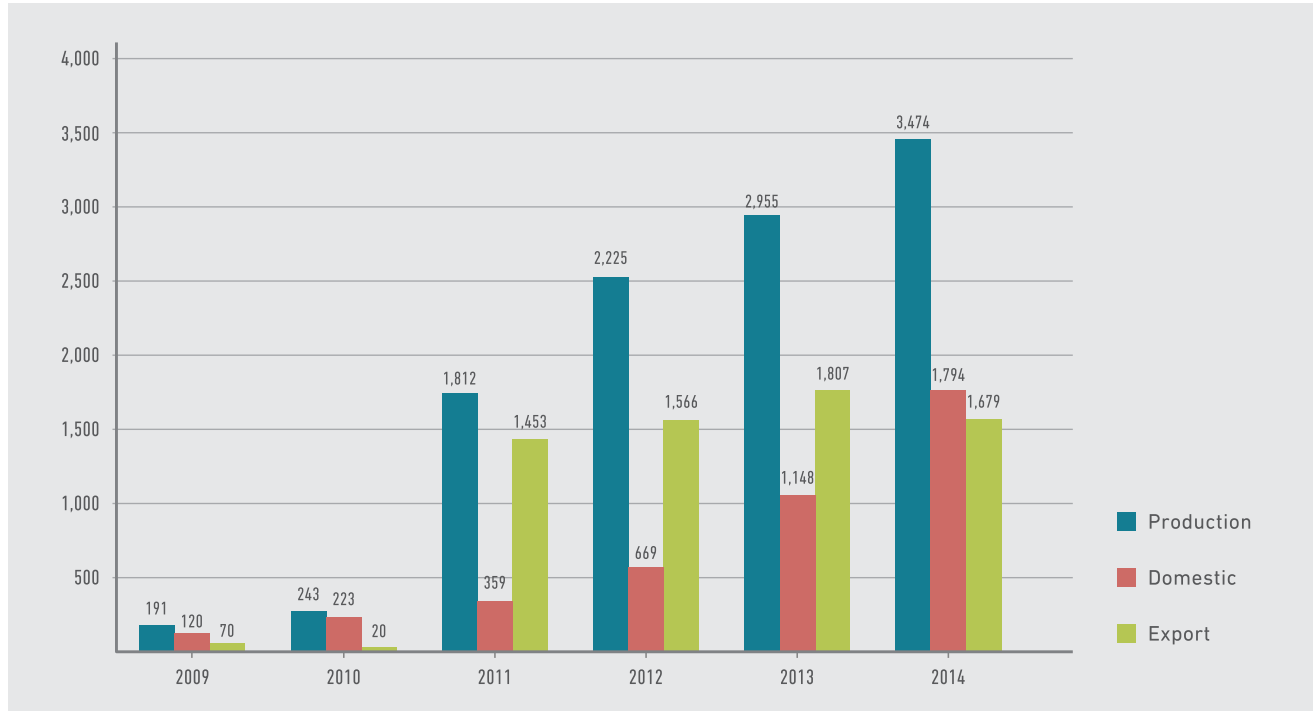
Source: Ministry of Energy and Mineral Resources, reprocessed by NEC, 2014

- Total hydropower potential 75 GW, installed capacity 7,097 MW (9.4% potential)
- Installed capacity data included minihydro and microhydro powerplant.

Total installed capacity increased by 19% when compared to 2013. Java is a region with the highest growth, by 15.7%

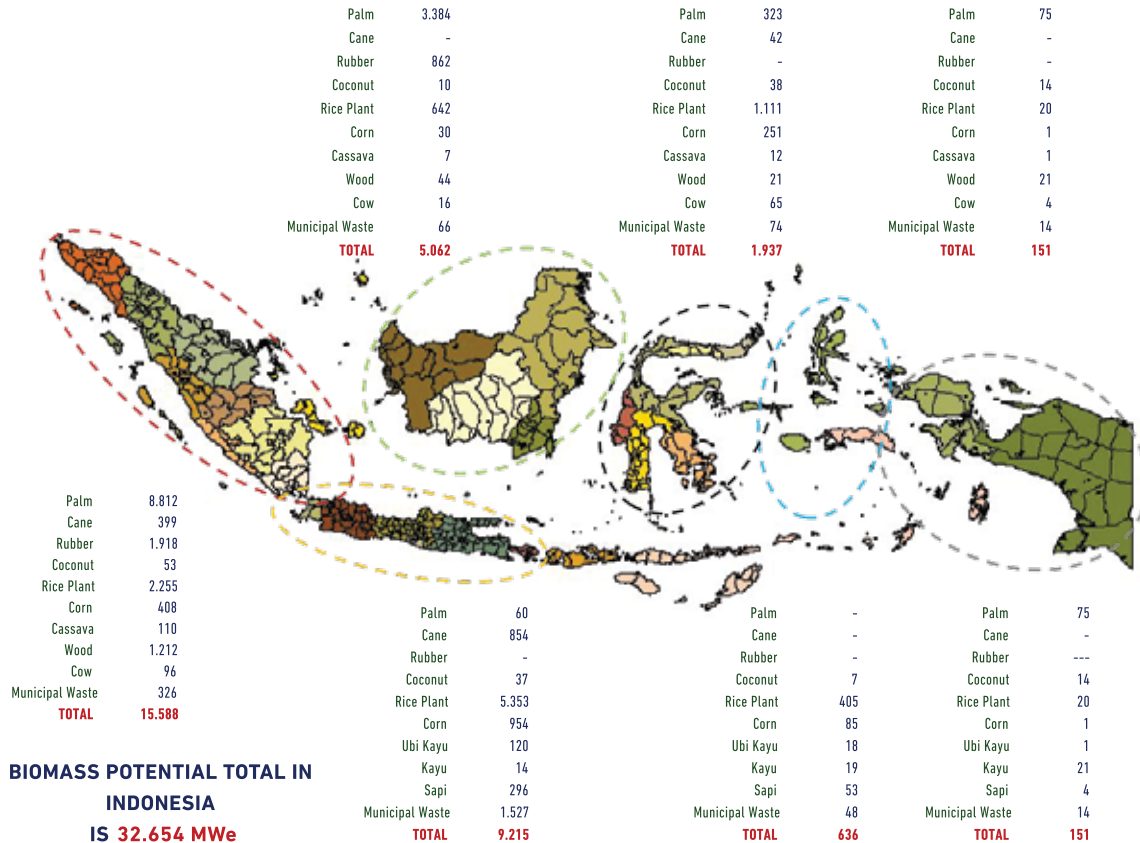
BIO-FUEL UTILIZATION

(Thousand KL)



Source: Ministry of Energy and Mineral Resources, reprocessed by NEC

NATIONAL BIOMASS POTENTIAL

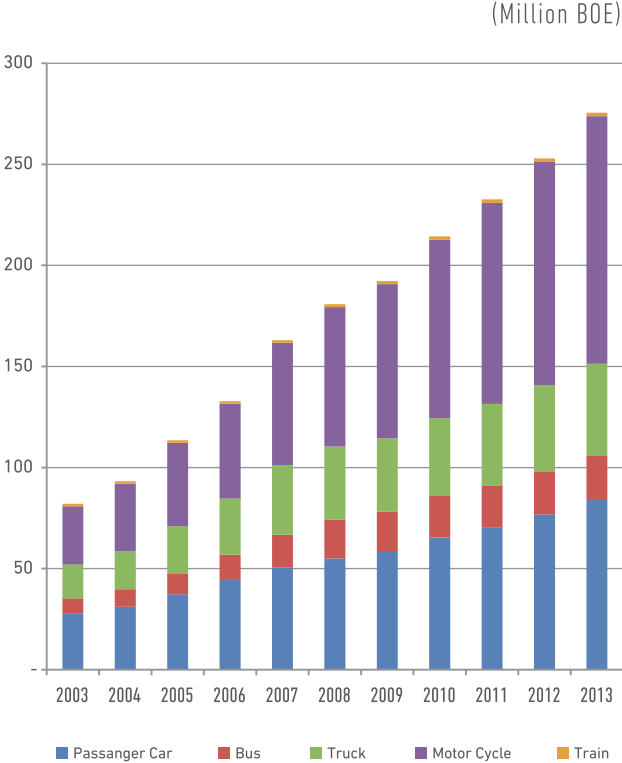
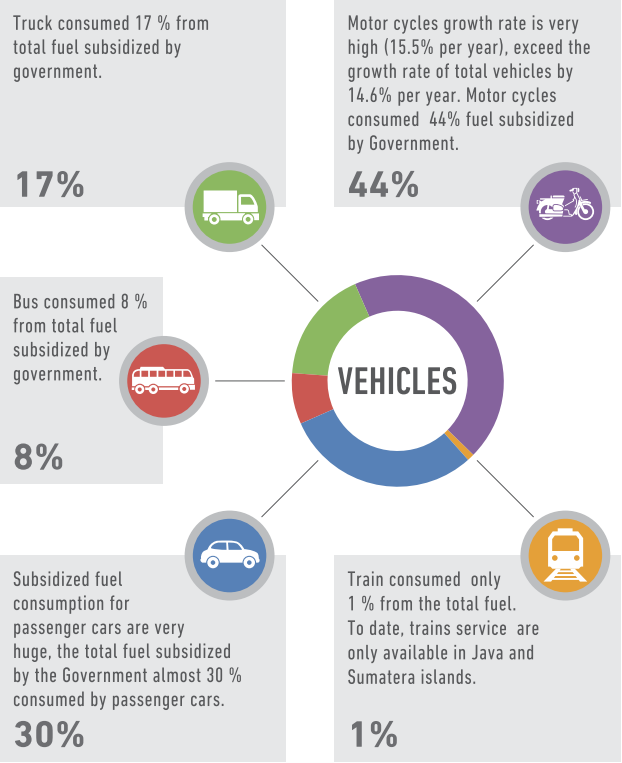




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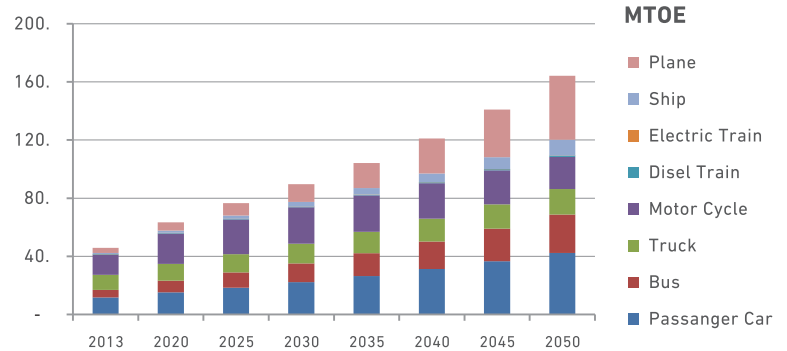
ENERGY CONSUMPTION BY SECTOR

ENERGY CONSUMPTION VEHICLES, 2013



TRANSPORTATION

Energy demand in transportation sector increasing around 3.5% per year, whereas in 2025, the energy needs reached 77 million TOE and increased to 164 million TOE in 2050. In total, the reduction in energy consumption in the transport sector due to National Energy Policy, around 11% in 2025 and 33% in 2050. The decrease is due to the transfer from personal transportation mode to the mass transportation, and also from technology efficiency and fuel substitution.



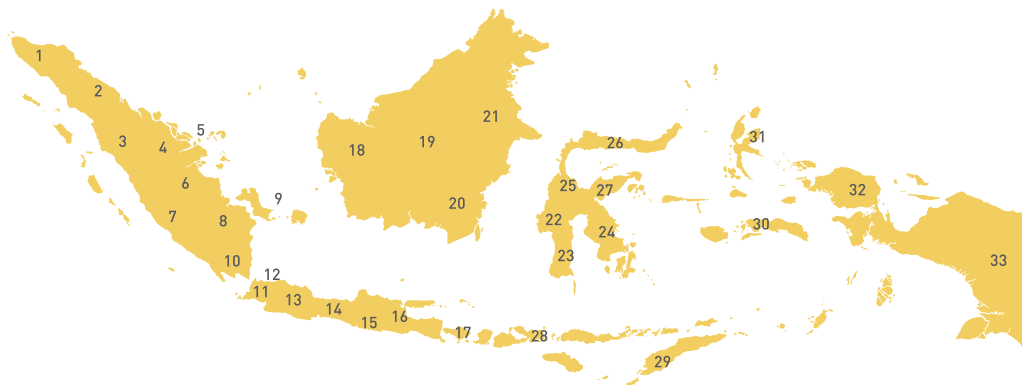
Transportation

					
	Train	Bus	Plane	MC	Ship
Capacity (People)	1,500	40	500	2	1,500
Consumption (Liter/Km)	3	0.5	40	0.08	10
Consumption (Liter/KM/People)	0.002	0.0125	0.05	0.04	0.006

The train advantages:

- Able to carry the passengers in large number
- Energy saving
- Land use saving
- Environmentally friendly
- high level safety
- Adaptive to technology developments
- Able to penetrate the heart of the city

ENERGY CONSUMPTION FOR INDUSTRIAL SECTOR



1. NAD	2. Sumatera Utara	3. Sumatera Barat	4. Riau	5. Kepulauan Riau	6. Jambi	7. Bengkulu	8. Sumatera Selatan	9. Bangka Belitung	10. Lampung	11. Banten	12. DKI Jakarta	13. Jawa Barat
17588	132431	29421	175787	231052	177926	6835	61765	416092	83057	417570	163327	819689
5	174913	1010841	286815	5460	68770	0	206000	62480	212653	646144	41164	1924763
101	14937	2	13503	5336	48	0	10836	7	202	53399	15551	82267
16118	455356	304363	103517	1035949	121565	2771	119278	6604	81474	2224937	1959680	7852385

14. Jawa Tengah	15. DI Yogyakarta	16. Jawa Timur	17. Bali	18. Kalimantan Barat	19. Kalimantan Tengah	20. Kalimantan Selatan	21. Kalimantan Timur	22. Sulawesi Barat	23. Sulawesi Selatan	24. Sulawesi Tenggara	25. Sulawesi Tengah	26. Sulawesi Utara
131883	25705	524510	9998	36848	72391	66992	70442	4071	39125	12678	5056	39125
530599	131604	663113	9371	17357	53	124757	42805	3	1254647	171477	1871	1254647
1347	11	153474	17	5	30	1	12	-	82	3	20	82
1775626	128090	3095922	21370	11718	3158	32071	90139	155	342012	879	5398	342012

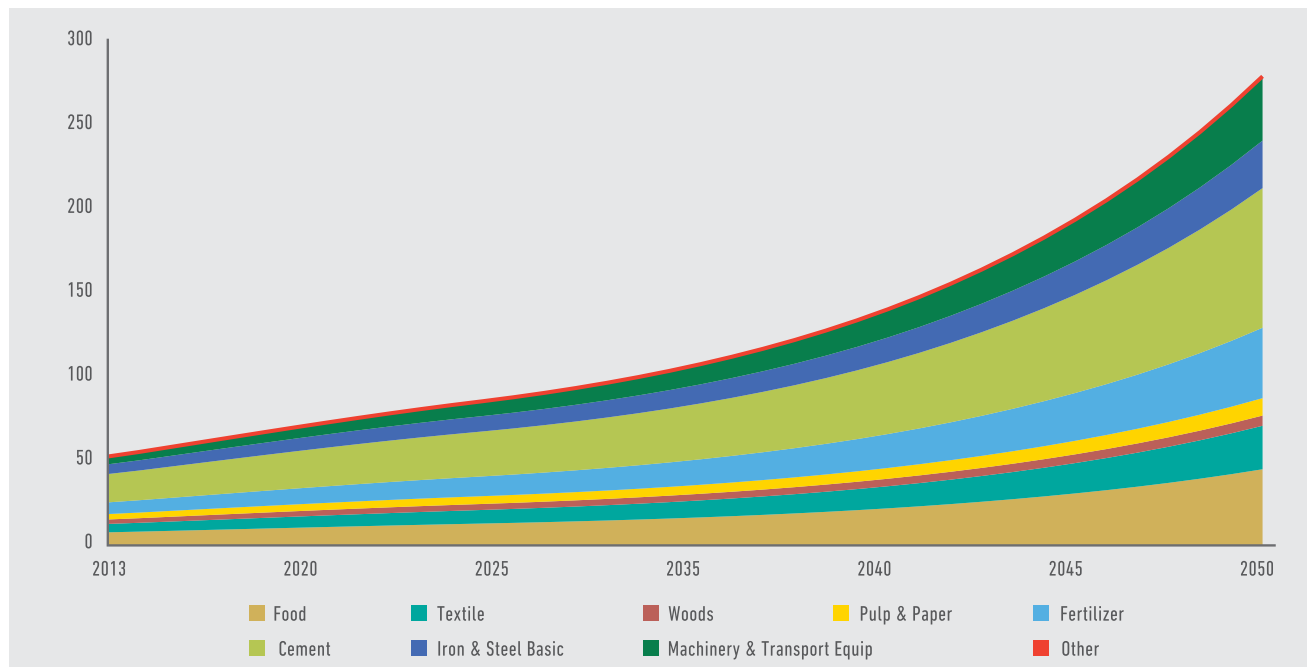
27. Gorontalo	28. Nusa Tenggara Barat (NTB)	29. Nusa Tenggara Timur (NTT)	30. Maluku	31. Maluku Utara	32. Papua Barat	33. Papua
918	1118	39125	6075	16	64810	15285
226	41	1254647	-	-	599	39
-	17	82	3	-	7042	29
261	1892	342012	977	54	6995	1333

Total

Fuel Oil (KiloLiter)	: 3,874,768
Coal (Ton)	: 7,648,460
Gas (Thousand M3)	: 358,288
Electricity (Thousand Kwh)	: 1,984,178

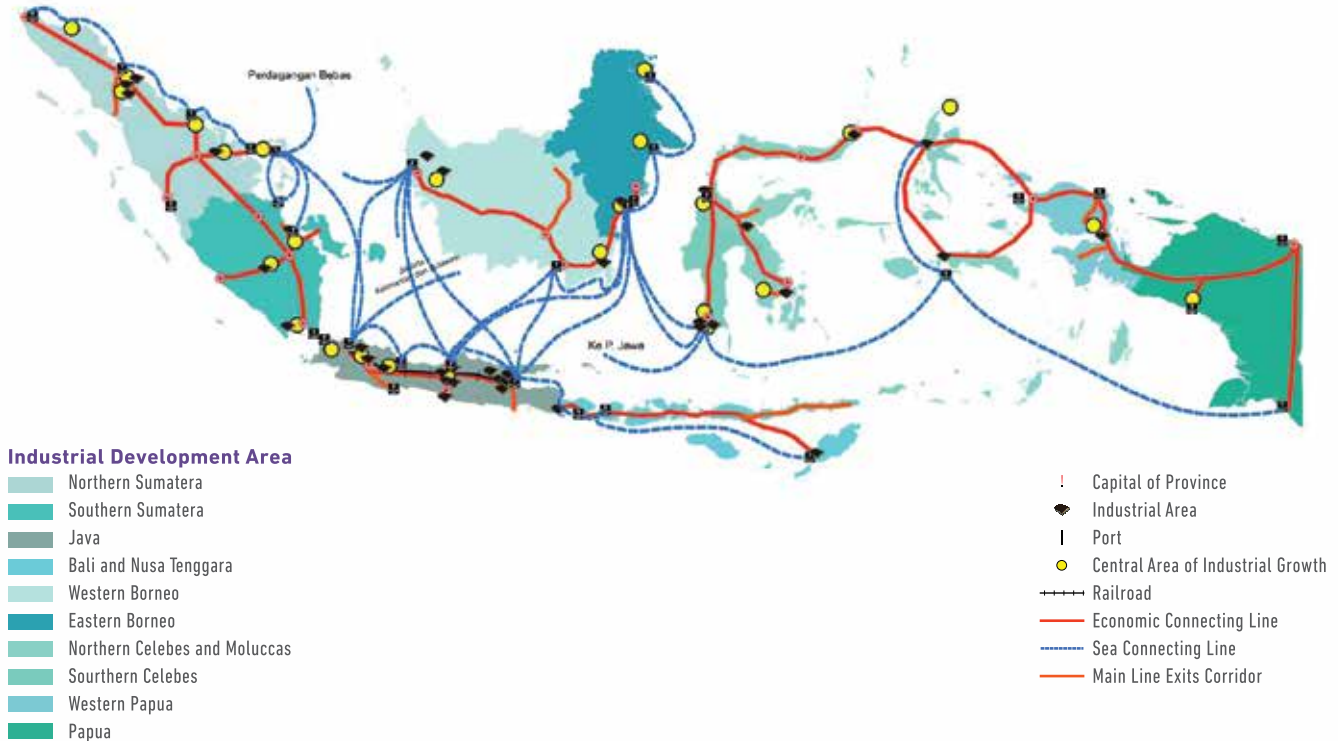
ENERGY DEMAND FOR INDUSTRIAL SECTOR

MTOE



Energy use in the industrial sector can be classified into three groups. First group - the largest energy user : cement and non metals (31-32%). second group - the middle group energy user : food and beverages, chemical fertilizer and rubber, metal and steel, and machinery and transportation equipment industries, each accounted for between 10-15% of energy. Third : low energy user group with a share of between 1-9%.

INDUSTRIAL DEVELOPMENT AREAS



POWER PLANT POTENTIAL BASED ON PALM OIL WASTE



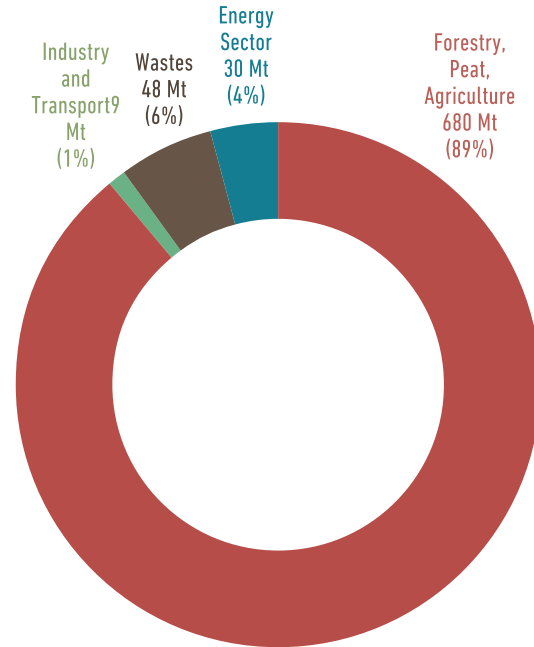
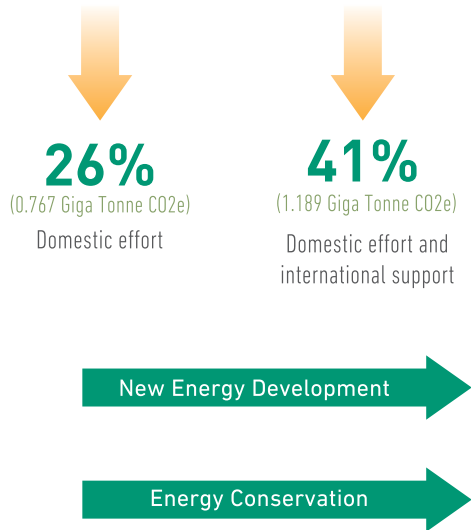


9 CO2 EMISSIONS

NATIONAL COMMITMENT TO REDUCE GHG

Presidential Regulation Number 61 Year 2011
to reduce greenhouse gasses (GHG) emission by 2020

(Indonesia Commitment in G-20 Pittsburgh and COP15)



Source: BP Statistical Review, reprocessed by NEC

EMISSION REDUCTION ROAD MAP IN ENERGY SECTOR

ACTION PLAN

Million Ton CO₂

2010-2014

5.06

- Energy Audit
- Energy Saving Lamps
- The provision and management of NRE
- Biogas utilization

- Gas for public transportation
- Increasing gas city connection to households
- Developing the LPG Mini Refinery Plant
- Reclamation for the ex-mining location

2015-2020

18.08

- Energy Audit
- Energy Saving Lamps
- The provision and management of NRE
- Biogas utilization

- Gas for public transportation
- Increasing gas city connection to households
- Developing the LPG Mini Refinery Plant
- Reclamation for the ex-mining location

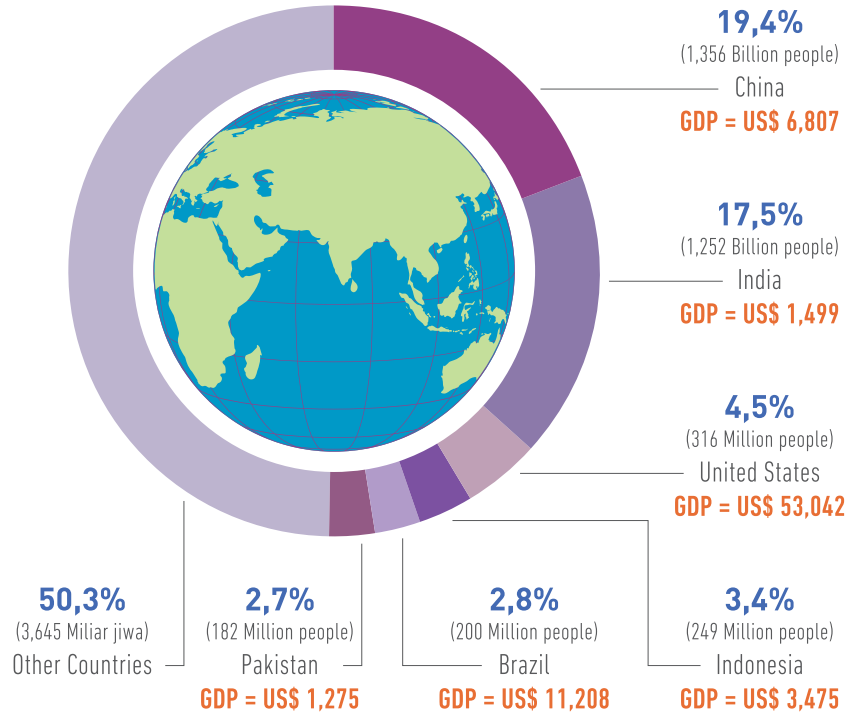
An abstract graphic in the top-left corner consisting of various blue, semi-transparent geometric shapes like cubes and rectangular prisms, some of which are stacked or overlapping, creating a sense of depth and movement.

10

**INDONESIA'S POSITION
IN THE GLOBAL CONTEXT**

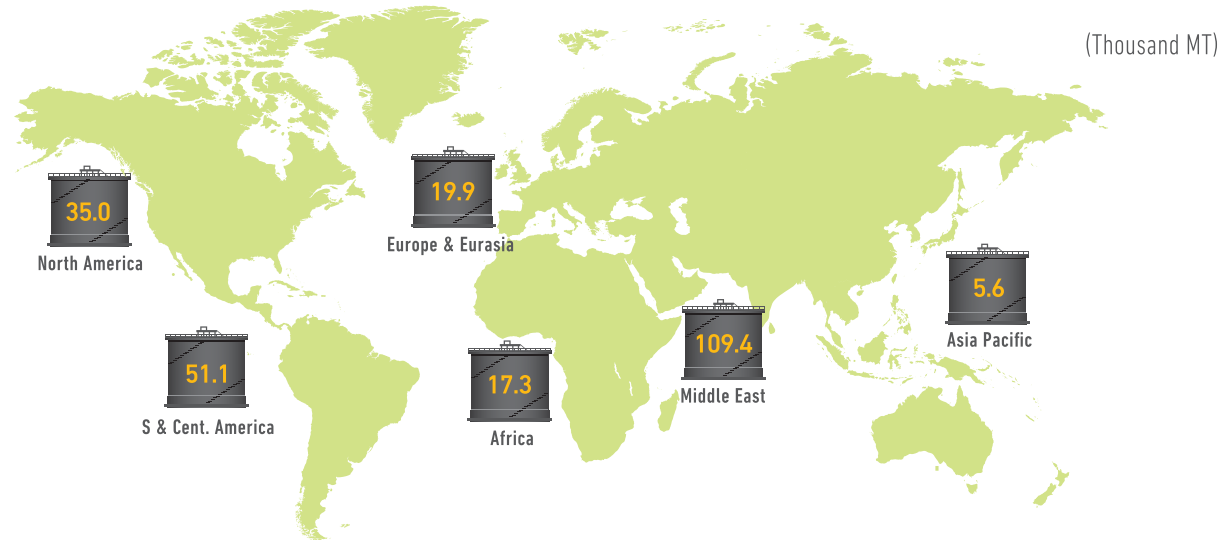
POPULATION AND GDP PER CAPITA 2013

- 19% of the people living in China
- Indonesia is the 4th largest population country (249 million people) with the GDP per capita amounted to 3,475 US. Dollars
- 18% of the world population has a high income, 12% low-income and the remaining are in the middle income.
- Luxembourg is the country with the largest GDP per capita is 110.697 US. Dollars with a population of only 543 Thousands peoples
- Malawi is a country with the lowest GDP is 226 US Dollars and total population of 16.363 Thousands people.



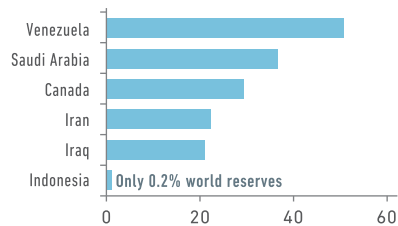
Source : World Bank December 2014, reprocessed by NEC

OIL, 2013

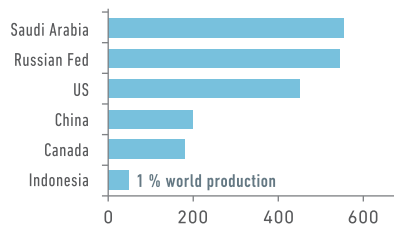


Source : BP Statistical Review 2014, reprocessed by NEC

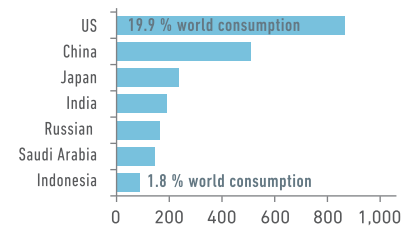
Proven Reserve in selected countries, 2013



Production in selected countries, 2013 (MT)

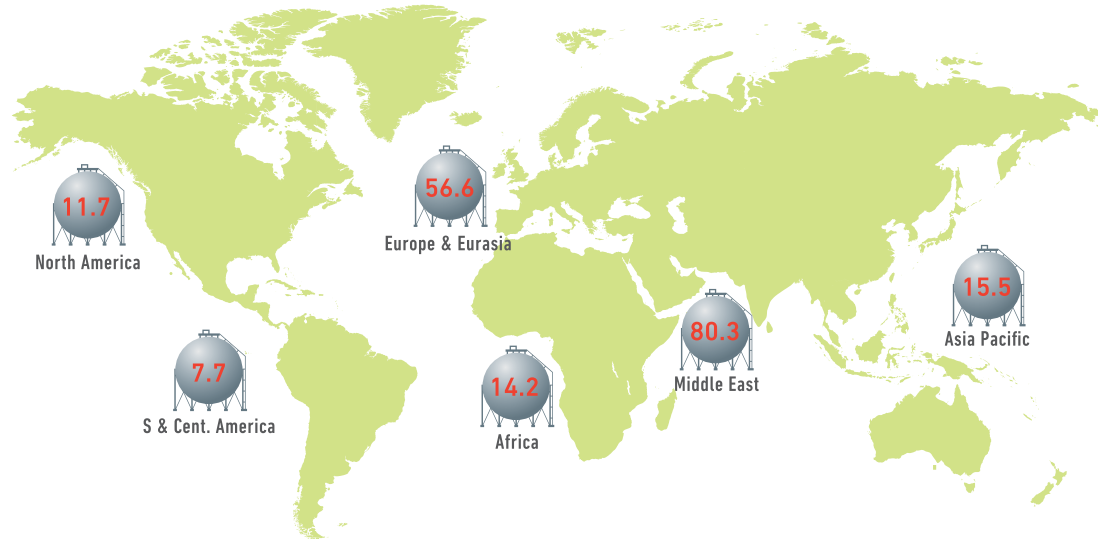


Consumption in selected countries, 2013 (MT)



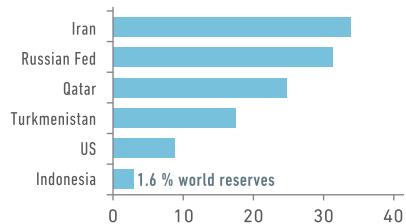
GAS, 2013

(TCM)

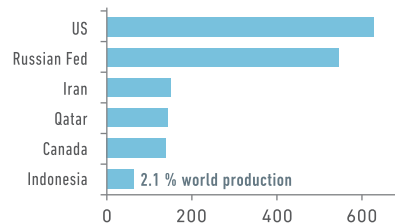


Source: BPS - Statistical Review 2014, reprocessed by NEC

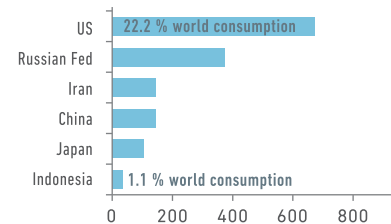
Proven Reserve in selected countries, 2013



Production in selected countries, 2013 (MTOE)



Consumption in selected countries, 2013 (MTOE)

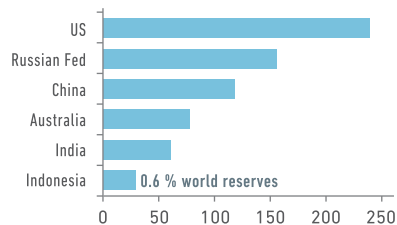


COAL

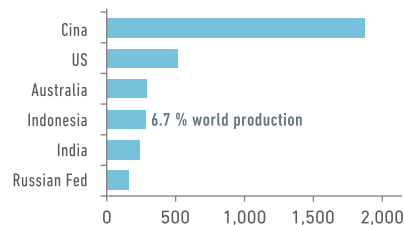


Source: BPS - Statistical Review 2014, reprocessed by NEC

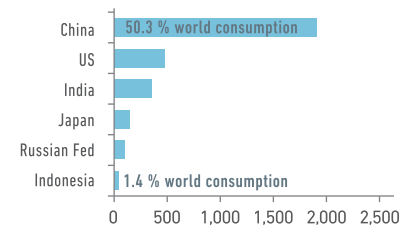
Proven Reserve in selected countries, 2013



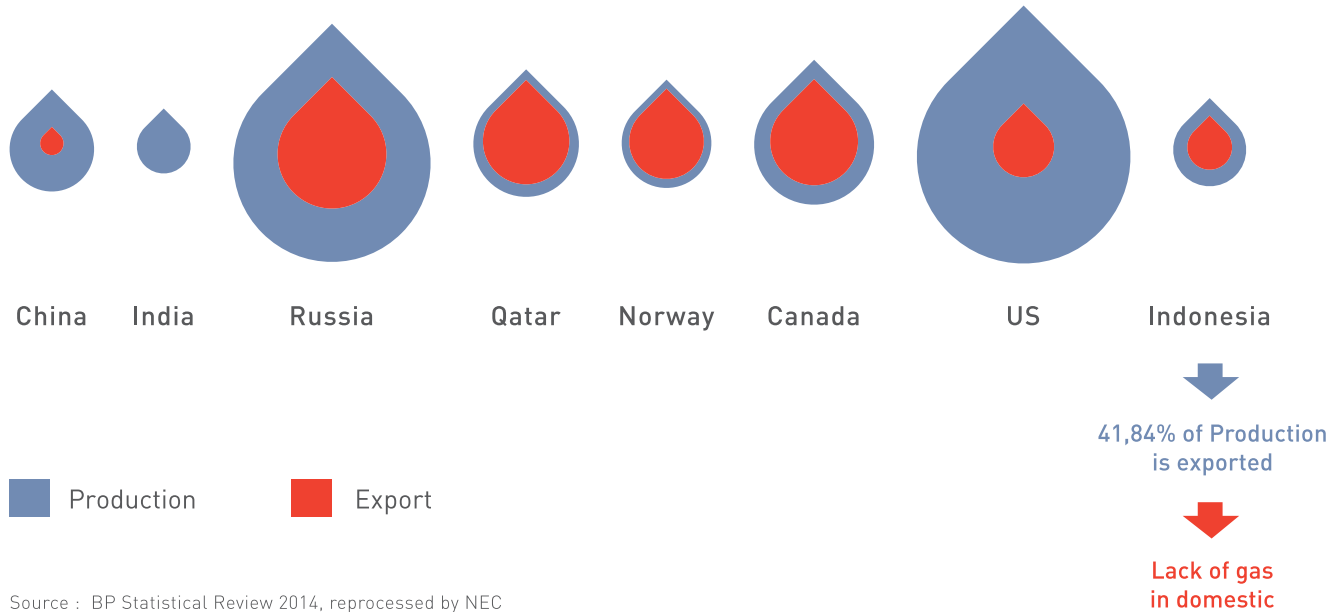
Production in selected countries, 2013 (MTOE)



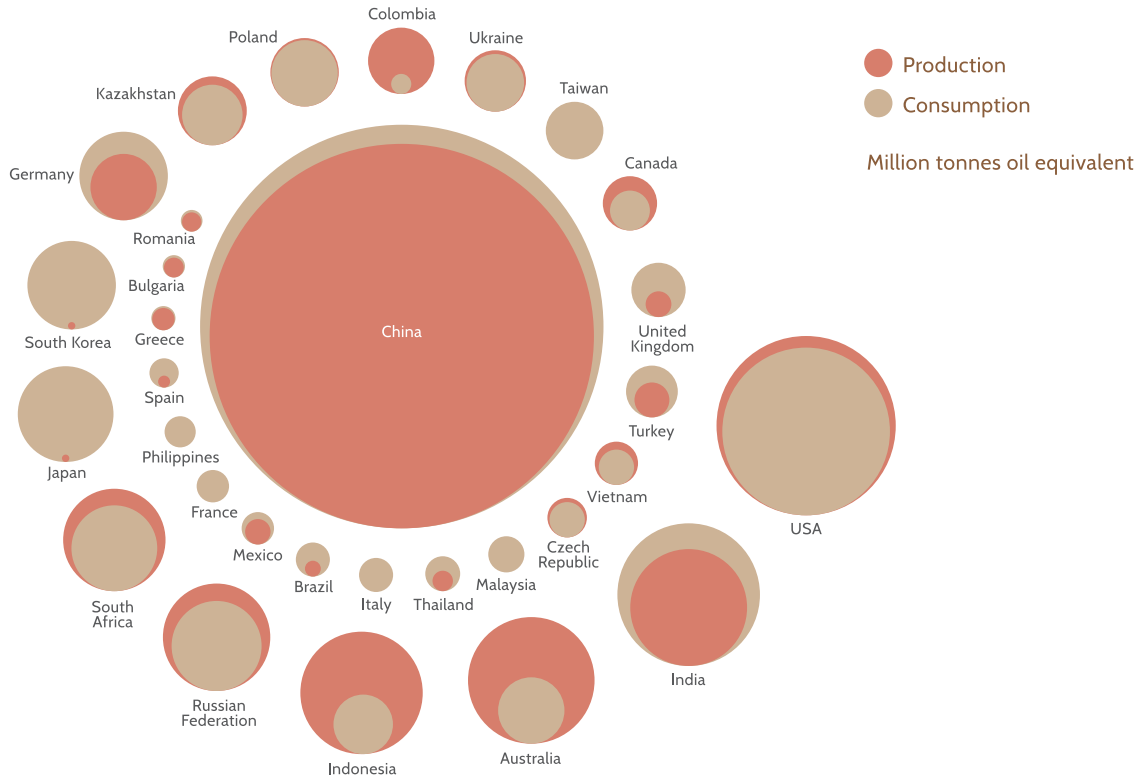
Consumption in selected countries, 2013 (MTOE)



GAS PRODUCTION vs EXPORT



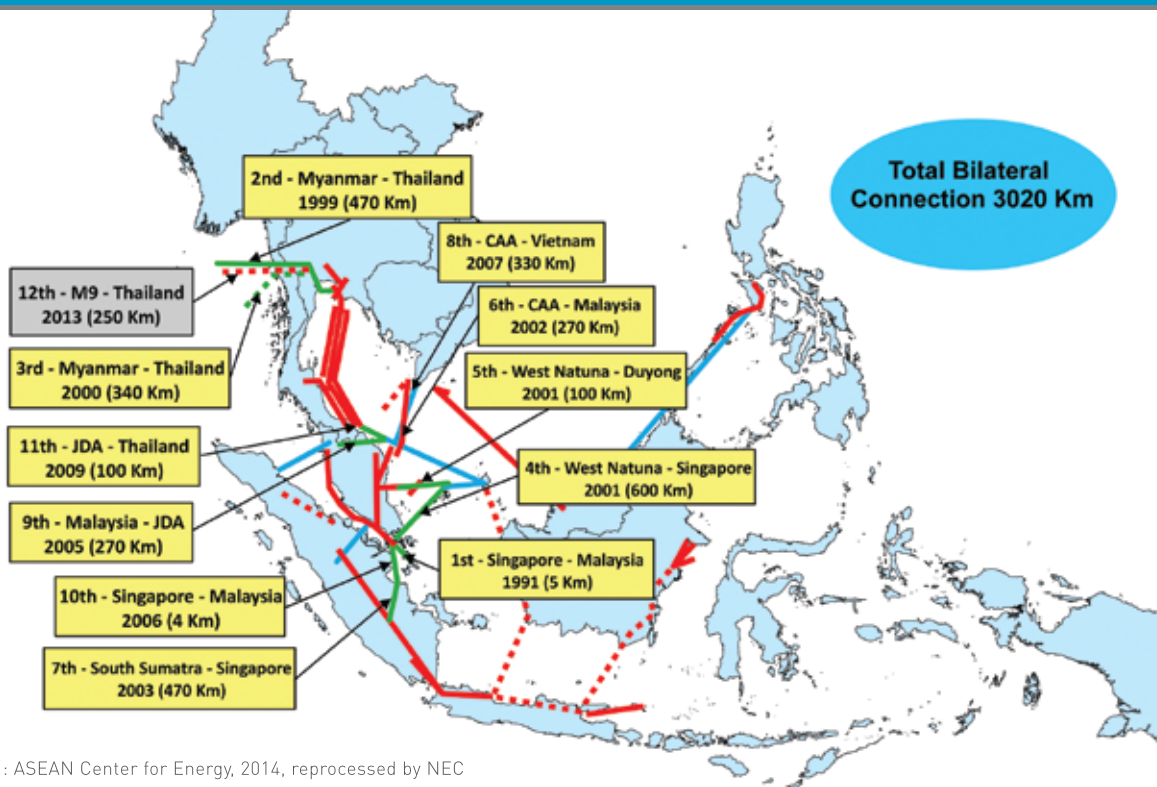
COAL PRODUCTION VS CONSUMPTION



Source : Wood Mackenzie Coal Supply Service, ANZ, reprocessed by NEC

GAS PIPELINES INFRASTRUKTUR IN ASEAN

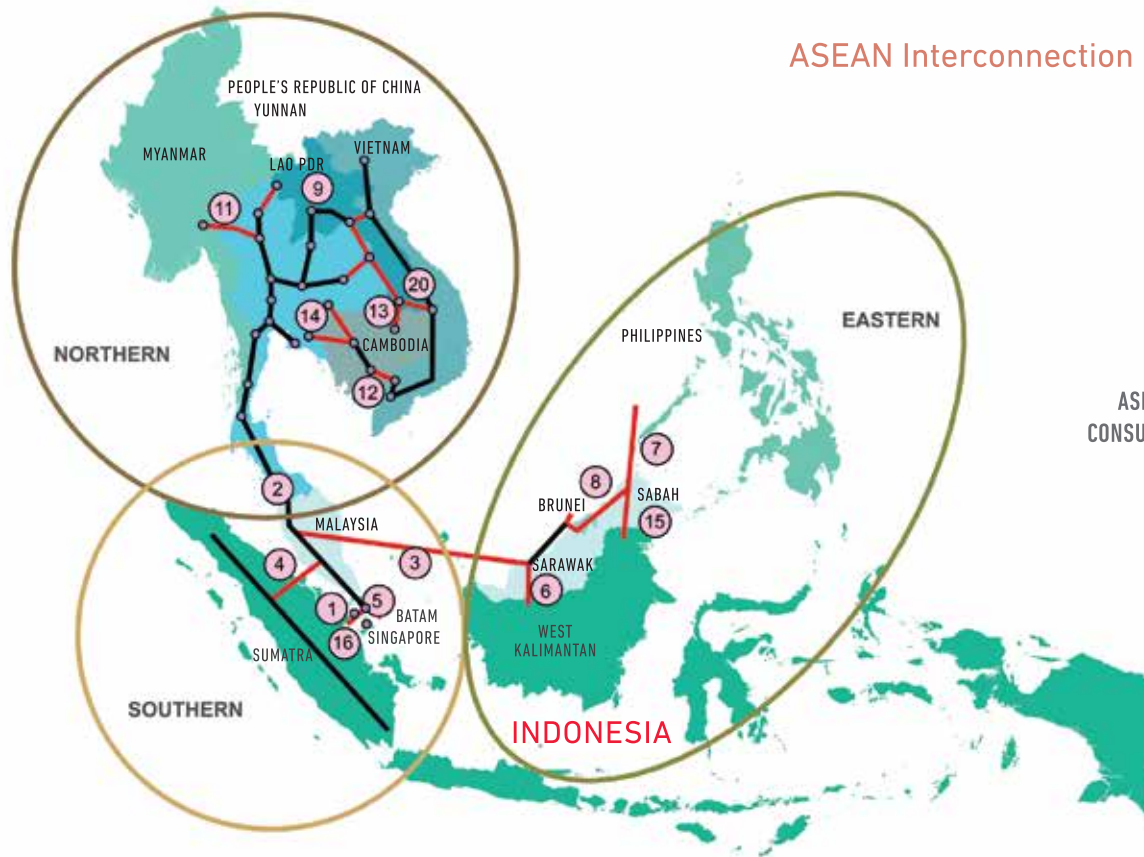
Development of cross border TAGP gas pipelines infrastructure in ASEAN (2008-2012)



Source : ASEAN Center for Energy, 2014, reprocessed by NEC

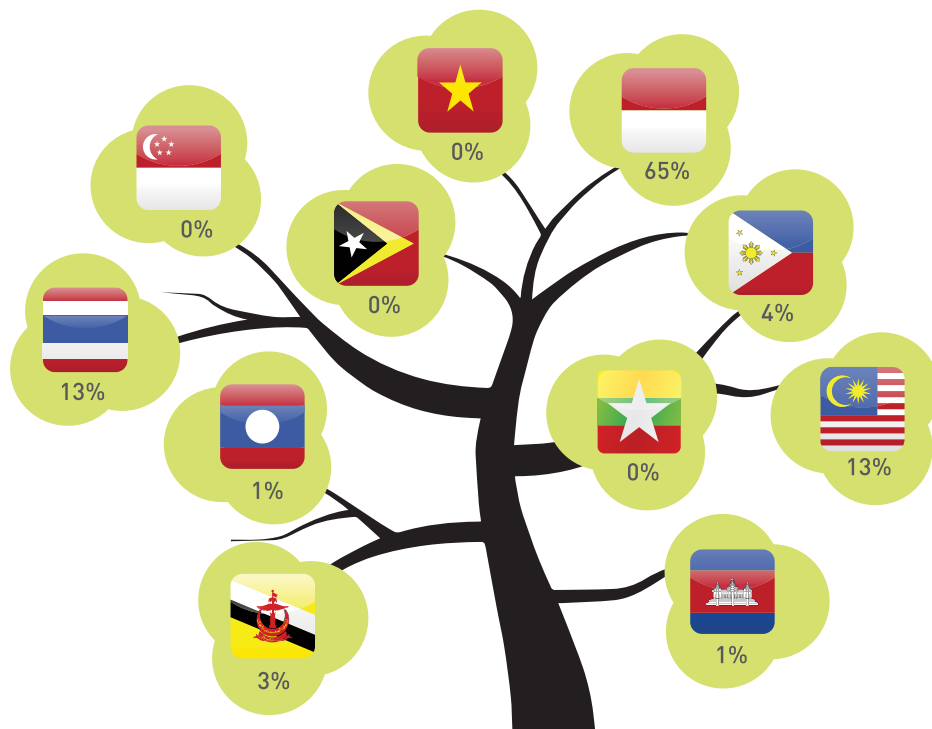
ASEAN POWER GRID

ASEAN Interconnection Sub-regions



ASEAN POWER GRID
CONSULTATIVE COMMITTEE

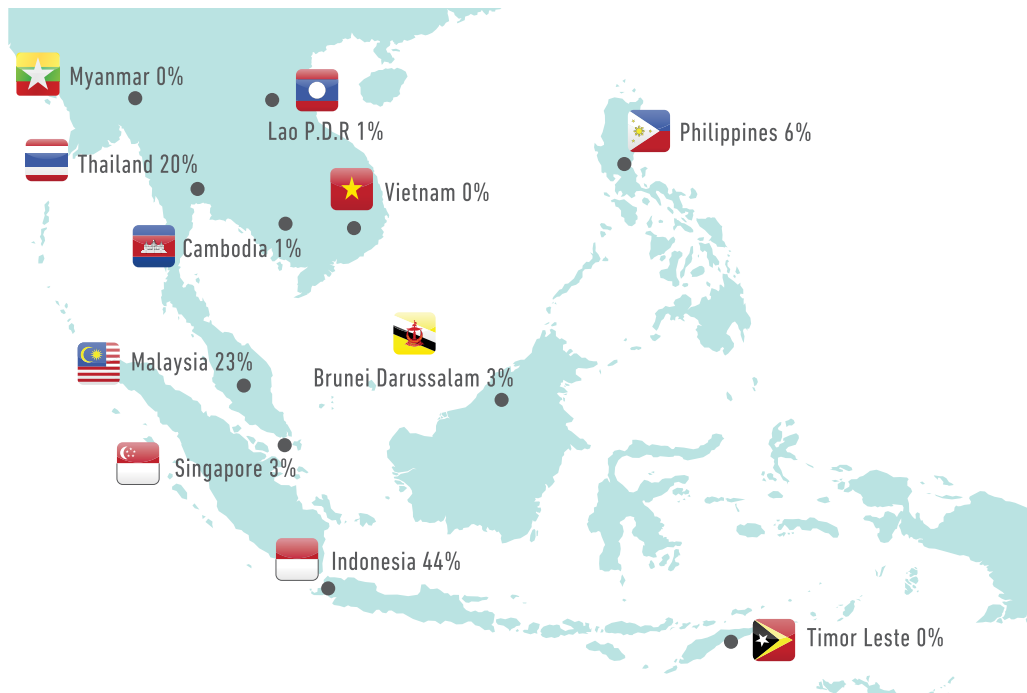
SHARE OF ASEAN PRIMARY ENERGY PRODUCTION



Source: ASEAN Center for Energy, 2011, reprocessed by NEC

Note : Data of Myanmar and Vietnam are not yet provided

SHARE OF ASEAN ENERGY DEMAND



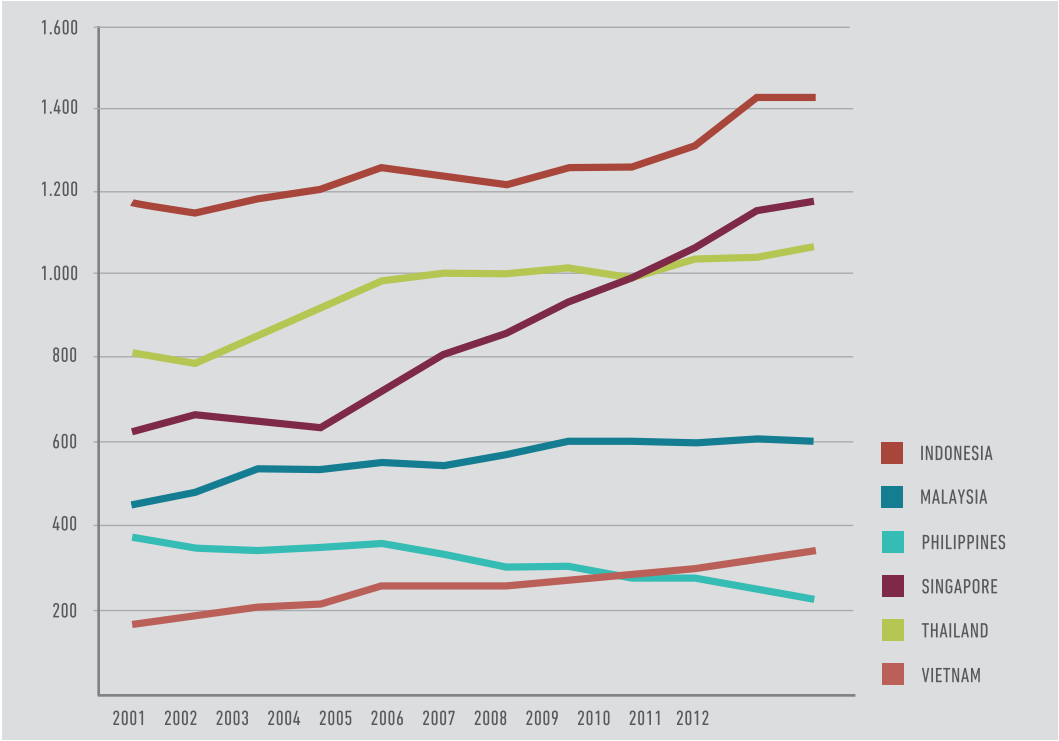
Oil demand rises from 4.4 mb/d today to 6.8 mb/d in 2035, natural gas demand increases by 80% to 250 bcm

Source : ASEAN Center for Energy, 2011, reprocessed by NEC

Note : Data of Myanmar and Vietnam are not yet provided

ASEAN OIL CONSUMPTION

(Thousand Barrel per Day)

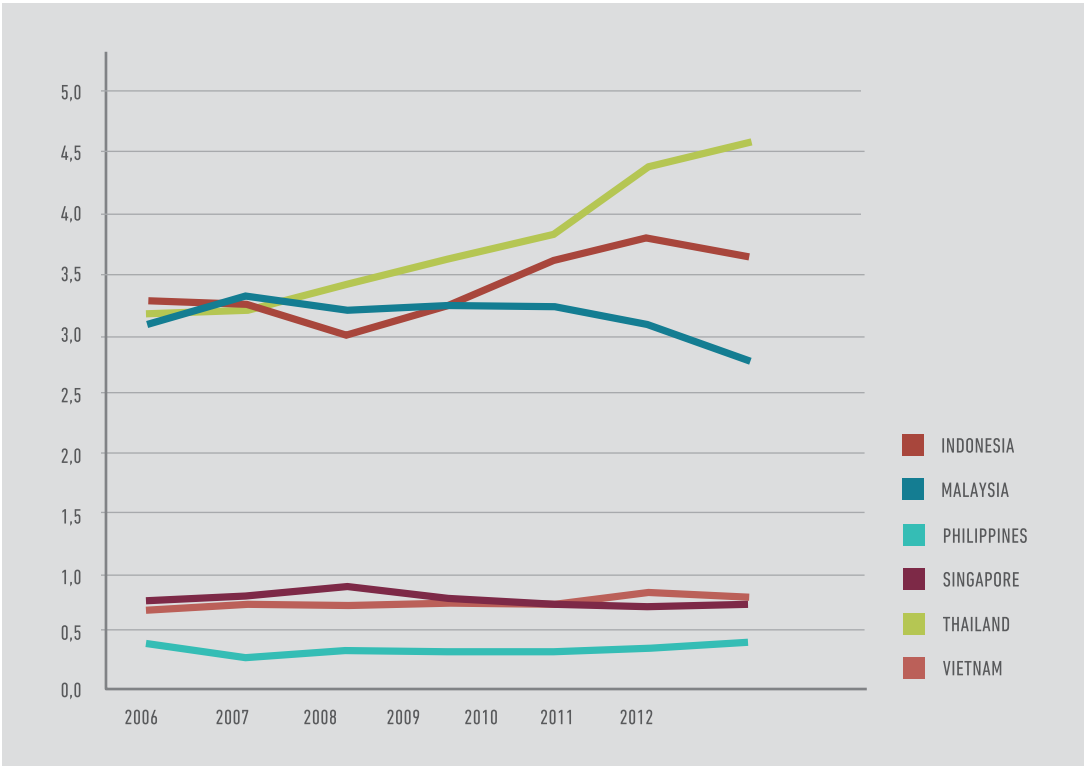


Source: ASEAN Center for Energy, reprocessed by NEC

Note : Data of other ASEAN member states are not yet provided

ASEAN OIL CONSUMPTION

(Thousand Barrel per Day)



Source: Data of other ASEAN member states are not yet provided

ABBREVIATION

Abbreviation	
Bbl	Barrel
BOE	Barrels of Oil Equivalent
BOEPD	Barrels of Oil Equivalent Per Day
GDP	Gross Domestic Product
GW	Giga Watt
GWh	Giga Watt Hour
KL	Kilo Litter
LPG	Liquefied Petroleum Gas
MBOE	Million Barrels of Oil Equivalent
MMSTB	Million Stock Tank Barrels
MMSCF	Million Standard Cubic Feet
MT	Million Ton
MTOE	Million Ton of Oil Equivalent
MW	Mega Watt
MWh	Mega Watt Hour
KL	Kilo Litter
KWh	Kilo Watt Hour



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